6 Crystal chemistry

6.1 Structures of metals, metalloids and their compounds[†]

The elements have been arranged in the following order:

1. Group Ia—Li, Na, K, Rb, Cs
2. Group IIa—Be, Mg, Ca, Sr, Ba
3. Group IIIa—Sc, Y, La, Ce, Pr, Nd, Eu, Gd, Tb, Dy, Ho, Er, Tm, Yb, Lu, Ac
4. Group IVa—Ti, Zr, Hf, Th, Pa, U, Np, Pu, Am, Cm
5. Group Va—V, Nb, Ta
6. Group VIa—Cr, Mo, W
7. Group VIIa—Mn, Tc, Re
8. Group VIII—Fe, Co, Ni; Ru, Rh, Pd, Os, Ir, Pt
9. Group Ib—Cu, Ag, Au
10. Group IIb—Zn, Cd, Hg
11. Group IIIb—Al, Ga, In, Tl
12. Group IVb—Si, Ge, Sn, Pb
13. Group Vb—As, Sb, Bi
14. Metalloids etc.—B, C, P, N, Te, Se, S

A compound or solid solution composed of the elements ABCD . . . is placed under that element, A, B, C, D, . . . which occurs *last* in the above list. If there are several compounds containing this particular element, they are arranged in the following order:

(a) Compounds containing the same elements—in the order of increasing content of the second element.
(b) Compounds of different elements—in the order in which the other elements occur in the above list.

For example, Na_2K is described under K, because K comes after Na in the list; MgSr and Mg_2Sr are both entered under Sr, and are described in that order, in accordance with (a); Li_2Ca and Mg_2Ca are entered under Ca, and described in that order, in accordance with (b).

The second column of Table 6.1 gives the symbol for the structural type to which the element or compound is assigned, the notation used being that of *Strukturbericht*.* Detailed descriptions of the structural types are given in Table 6.2.

The third column of Table 6.1 gives the lattice constants of the various elements and compounds. For cubic crystals the single parameter a is given, in Å units; for tetragonal and hexagonal crystals a and c, in that order; for orthorhombic, a, b and c; for rhombohedral a, α; for monoclinic a, b, c, β.

Temperatures given in parentheses are those at which allotropic or polymorphic modifications are stable; figures such as $A = 28, M = 4$, also given in parentheses, refer to the number of atoms (A) or molecules (M) included in the unit cell; alternative structures are given where the available evidence is insufficient to permit of a decision being reached between them, and in such cases the authorities are quoted. Finally, it should be noted that in the case of the sulphides, only those having relatively simple structures have been included.

[†] For information on crystallography, see Chapter 5. Techniques that enable determination of crystal structure can be found in Chapter 4 and Section 10.4.

* Strukturbericht of *Z. Krystallographie*, Leipzig.

Although this system of classification is not used in any other compilation of intermetallic compounds, it has been retained as it has its own logic in keeping together compounds of similar systems in a Group; this would be lacking in a strictly alphabetical classification.

Since this compilation was first made, data on a very large number of intermetallic compounds have been published, far more than can be included in this list without extending it to over 1000 pages. Table 6.1 refers to the more frequently encountered compounds. Other compilations should be consulted for compounds not found in Table 6.1.

A comprehensive database is: 'Pearson's Handbook of Crystallographic Data for Intermetallic Phases' (4 volumes) by P. Villars and L. D. Calvert, published by A.S.M., Materials Park, OH, 1991. For an annually updated database, see the 'Powder Diffraction File', published by the International Center for Diffraction Data, Newtown Square, PA.

Pearson has introduced a new system of symbols based on the crystallography of the classes of isostructural compounds. This cannot be deduced readily from the Structurbericht Classification used in Table 6.1 and described in Table 6.2. Hence a comparison of the two nomenclatures is provided in Table 6.3.

The Pearson symbols take the form aBn, where:

a denotes the crystal system, i.e. cubic, hexagonal, etc.

B denotes the space lattice of the unit cells using the Hermann–Mauguin symbols, i.e. P, C, F, I, etc.

n denotes the number of atoms per unit cell.

The crystal systems are:

c = Cubic o = Orthorhombic
h = Hexagonal t = Tetragonal
m = Monoclinic

The space lattices of the unit cells are:

P = Simple C = One face-centred*
F = All Face-centred+ R = Simple (rhombohedral)
I = Body-centred

*Monoclinic and Orthorhombic
+Cubic and Orthorhombic
e.g. A1(Cu) is cF4 and C14($MgZn_2$) is hP12.

As will be evident from Table 6.3, a Pearson symbol can be the same for several different structural types or prototypes. See note at end of references for sources of data.

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Group Ia: Li, Na, K, Rb, Cs			
Li	*A*2	3.509 3	1
	*A*3	3.111; 5.093 (spontaneous transformation at −195°C)	
	*A*1	4.355 (high pressure phase)	
Na	*A*2	4.282 0(5)	
	*A*3	3.767; 6.154 (low temperature phase below −237°C)	
K	*A*2	5.330 17(34)	
Na_2K	*C*14	7.48; 12.27	
Rb	*A*2	5.585	
Cs	*A*2	6.141(7)	
Group IIa: Be, Mg, Ca, Sr, Ba			
Be	*A*3	2.282 6; 3.583 6	
$Be_{13}Mg$	$D2_3$	10.166	2
Ca (α)	*A*1	5.588 4 (<~443°C)	
(γ)	*A*3	4.00(2); 6.50(2) (>~443°C)	
Li_2Ca	*C*14	6.248(8); 10.23(2)	
$Be_{13}Ca$	$D2_3$	10.312	2
Mg_2Ca	*C*14	6.225; 10.18	

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Group IIa: Be, Mg, Ca, Sr, Ba—*continued*			
Sr	*A1*	6.084 9(5)	
MgSr	*B2*	3.900	
Mg_2Sr	*C15*	6.426(8); 10.473(8)	
Ba	*A2*	5.013(5)	3
Mg_2Ba	*C15*	6.636(8); 10.655(8)	
Group IIIa: Sc, Y and Rare Earths			
Sc (α)	*A3*	3.310(1); 5.274(2)	
Y	*A3*	3.651 5(2); 5.747 4(4)	
$Be_{13}Y$	$D2_3$	10.24	
La (α)	*A3*	3.770; 12.159 (room temp.)	4
(β)	*A1*	5.31 (above 310°C)	4
MgLa	*B2*	3.97	
Mg_2La	*C15*	8.774	
Mg_3La	$D0_3$	7.478	
Ce (α)	*A1*	4.832(4)	
(β)	*A3*	3.16;	4
$Be_{13}Ce$	$D2_3$	10.376	
MgCe	*B2*	3.90	
Mg_2Ce	*C15*	8.733	
Mg_3Ce	$D0_3$	7.42	
Pr	*A1*	5.183	
	A3	3.671 5; 11.830	4
MgPr	*B2*	3.88	
Mg_3Pr	$D0_2$	7.42	
Nd	*A3*	3.656; 11.798	207
Eu	*A2*	4.582 7	208
Gd	*A3*	3.633 6; 5.781	209
Tb		3.605 5; 5.696 6	
Dy		3.591 5; 5.650 1	
Ho		3.576 4; 5.613 6	
Er		3.561 0(2); 5.592 9(10)	210
Tm		3.537 5; 5.554	
Yb	*A1*	5.484 7	
Lu	*A3*	3.503 1(4); 5.550 9(4)	
Ac	*A1*	5.670	6
Group IVa: Ti, Zr, Hf, Th and Pa, U, Np, Pu, Am, Cm			
Ti (α)	*A3*	2.950 3(6); 4.681 0(2) (<~882°C)	
(β)	*A2*	3.311 2 (>~882°C)	
$BeLi_4$	hex.	10.92; 8.94 ($M = 6$)	211
Be_2Ti	*C15*	6.453 2(4)	
$Be_{12}Ti$	*D2a*	29.44; 7.33 (may be considered as *D2b* 7.35; 4.19)	
Ti_2Be_{17}	Th_2Ni_{17}	7.36; 7.30	
Ti_2Be_{17}	Nb_2Be_{17}	7.34; 10.73	
Zr (α)	*A3*	3.231 78; 5.148 31 (<~863°C)	7
(β)	*A2*	3.568(5) (>~863°C)	7
Be_5Zr	*D2d*	4.564(2); 3.485(2)	
$Be_{17}Zr_2$	$Be_{17}Nb_2$	7.538 1(1); 11.015(1)	
Be_2Zr	*C32*	3.814; 3.230	8
$Be_{13}Zr$	$D2_3$	10.044	
Hf (α)	*A3*	3.198; 5.061 < 1743°C	9, 10
(β)	*A2*	3.615	
Hf_2Be_{17}	Nb_2Be_{17}	7.494; 10.93	213
$HfBe_{13}$	$D2_3$	10.005	
$HfBe_5$	*D2d*	4.519; 3.465	
Th	*A1*	5.084 2	
$B_{13}Th$	$D2_3$	10.41	
Mg_2Th	*C36*	6.086; 19.64 (at and below 700°C)	12
	C15	8.570 (at and above 800°C)	12
Pa	*A6*	3.940; 3.244	13

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Group IVa: Ti, Zr, Hf, Th and Pa, U, Np, Pu, Am, Cm—*continued*			
U (α)	*A*20	2.853 7; 5.869 5; 4.954 8	214
(β)	$D8_6$	10.758 9; 5.653 1 (range 668–776°C)	14
(γ)	*A*2	3.533 5 (>776°C)	
UH_3	*A*15	4.147(3)	
$Be_{13}U$	$D2_3$	10.248 87(2)	
TiU_2	*C*32	4.828; 2.847	16
Np (α)	*Ac*	(<278°C) 6.663; 4.723; 4.887	
(β)	*Ad*	4.897(2); 3.388(2) (range 278–540°C)	
$B_{13}Np$	$D2_3$	10.266 to 10.256	19
Pu (α)	monocl.	6.183; 4.822; 10.963; $\beta = 101.79$	20
(β)	monocl.	11.830; 10.449; 9.227; $\beta = 138.65$	215
(δ)	*A*1	4.634 7	21
(δ')	*A*6	3.339(3); 4.446(7)	21
(ε)	*A*2	3.637 5	21
Am	$P6_3/mmc$	3.468 1(2); 11.241(3)	216
Group Va: V, Nb, Ta			
V	*A*2	3.027	22
Be_2V	*C*14	4.385; 7.130	
$Be_{12}V$	*D*2*b*	7.35; 4.24	
Be_2Nb_3	*D*5*a*	6.49(1); 3.35(1)	217
$Be_{17}Nb_2$	$R\bar{3}m$	7.409(2); 10.84(1)	218
$Be_{12}Nb$	*D*2*b*	7.376(5); 4.258(5)	
Nb	*A*2	3.300	
Ta	*A*2	3.302 56(5)	
$BeTa_2$	*C*16	6.010(4); 4.890(4)	
$Be_{17}Ta_2$	$R\bar{3}m$	7.388; 10.74	
Be_2Ta	*C*15	6.507	
ZrV_3	*C*14	5.28; 8.65	
TiV	*A*2	3.159	219
Group VIa: Cr, Mo, W			
Cr	*A*2	2.884 4	
CrH	*B*1	3.860 5(5)	
Be_2Cr	*C*14	4.260; 6.988	
$Be_{12}Cr$	*D*2*b*	7.238(4); 4.174(2)	
TiCr	*A*2	3.12	219
$TiCr_2$	*C*15	6.932(4)	
$ZrCr_2$ (1)	*C*14	5.102; 8.273	24
(2)	*C*15	7.205	24
$NbCr_2$	*C*15	6.991	
$TaCr_2$ (1)	*C*15	6.985	
(2)	*C*14	4.950; 8.245	
Mo	*A*2	3.147 0	
Be_2Mo	*C*14	4.448; 7.310	
$Be_{12}Mo$	*D*2*b*	7.251(2); 4.234(1)	25
$Be_{13}Mo$	tetr.	10.27; 4.29 ($M = 4$)	
$Be_{22}Mo$	cubic f.	11.634(2)	
$BeMo_3$	*A*15	4.89	220
$ZrMo_2$	*C*15	7.587 5	26
U_2Mo (γ)	*C*11*b*	3.427; 9.834	27
W (α)	*A*2	3.163 79	
Be_2W	*C*14	4.559; 7.333	
WBe_{22}	O_h^7	11.631(2)	220
$Be_{12}W$	*D*2*b*	7.362(5); 4.216(5)	
ZrW_2	*C*15	7.613(2)	
Group VIIa: Mn, Tc, Re			
Mn (α)	*A*12	8.912 5 (<727°C)	
(β)	*A*13	6.315 2 (727–1100°C)	28
(γ)	*A*1	3.860 (1100–1138°C)	
(δ)	*A*2	3.080 (>1138°C)	30

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Group VIIa: Mn, Tc, Re—*continued*			
YMn_2	*C*15	7.675 ($M = 8$)	221
YMn_4	bc.tetr.	8.808; 12.521 ($M = 12$)	222
YMn_{12}	*D*2*b*	8.541; 4.785	
Be_2Mn	*C*14	4.231; 6.909	
$Be_{12}Mn$	*D*2*b*	7.276(5); 4.256(5)	
$GdMn_2$	*C*15	7.74	
$TiMn_2$	*C*14	4.832; 7.930	31
TiMn	*D*8*b*	8.880; 4.542	
$ZrMn_2$	*C*14	5.026; 8.258	
$ThMn_{12}$	*D*2*b*	8.74(1); 4.95(1)	
Th_6Mn_{23}	*D*8*a*	12.443	
$ThMn_2$	*C*14	5.485; 8.896	
UMn_2	*C*15	7.162 8	
U_6Mn	*D*2*c*	10.312(5); 5.255(4)	
VMn_4	*D*8*b*	8.92; 4.61	
$NbMn_2$	*C*14	4.872; 7.975	
$TaMn_2$	*C*14	4.864; 7.943	
$CrMn_3$	*D*8*b*	8.86; 4.59	
Tc	*A*3	2.740 7(1); 4.398 0(1)	
Re	*A*3	2.761 5; 4.456 6	
Be_2Re	*C*14	4.345; 7.087	
$Be_{22}Re$	O_h^7	11.560(1)	
$ZrRe_2$	*C*14	5.259; 8.615	
$HfRe_2$	*C*14	5.254(1); 8.600(2)	223
URe_2	ortho.	<180°. 5.600; 9.180; 8.460	
	*C*14	>180°. 5.405; 8.683	224
Group VIII: Fe, Co, Ni; Ru, Rh, Pd; Os, Ir, Pt			
Fe (α)	*A*2	2.867 0; (<912°C and >1394°C)	
(γ)	*A*1	3.654 4 (912–1394°C)	
Be_2Fe	*C*14	4.215; 6.853	
Be_5Fe	*C*15	5.875 (stable only > ~1000°)	
Be_7Fe	hex.	4.137(2); 10.720(5)	
$Be_{12}Fe$	*D*2*b*	7.253(5); 4.232(5)	
$CeFe_2$	*C*15	7.300	
Gd_2Fe_3	cubic	8.25 ($M = 6$)	
$GdFe_3$	rhomb.	5.157; 24.70; 120°,	
Gd_2Fe_7	ortho.	5.71; 6.78; 7.15 ($M = 2$)	
$GdFe_4$	hex.	5.15; 6.64 ($M = 2$)	225
$GdFe_5$	ch	4.92; 4.11 ($M = 1$)	
Gd_2Fe_{17}	hex.	8.496; 8.345	
$GdFe_2$	*C*15	7.39	
YFe_2	*C*15	7.37	221
$TiFe_2$	*C*14	4.79; 7.81	
TiFe	*B*2	2.976	
$TiFe_3O$	$E9_3$	11.15	
Ti_4Fe_2O	$E9_3$	11.28	32
Ti_2Fe	$E9_2$	11.305	33
$ZrFe_2$	*C*15	7.065	
$ZrFe_2$	*C*36	4.988; 16.32	
$ThFe_5$	*D*2*d*	5.111 3(4); 4.052 2(4)	
U_6Fe	*D*2*c*	10.302 2; 5.238 6(1)	
UFe_2	*C*15	7.058	226
VFe	*D*8*b*	8.965; 4.633	
$NbFe_2$	*C*14	4.845; 7.893	
$TaFe_2$	*C*14	4.825; 7.875	
CrFe (σ)	*D*8*b*	8.796 6(6); 4.558 2(3)	35
MoFe (σ)	*D*8*b*	9.190; 4.814	35
Mo_6Fe_7	$D8_5$	4.754 6(5); 25.716(3); 120°	
WFe_2	*C*14	4.737(3); 7.694(4)	
W_6Fe_7	$D8_5$	4.764(3); 25.85(2); 120°	

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Group VIII: Fe, Co, Ni; Ru, Rh, Pd; Os, Ir, Pt—*continued*			
$MnFe_4$	*A*3	2.530; 4.079 (not an equilibrium phase)	
Co (α)	*A*3	2.507 1(3); 4.069 5(5)	227
(β)	*A*1	3.568 8 (>422°C)	
BeCo	*B*2	2.624	
$Be_{12}Co$	*D*2*b*	7.237(5); 4.249(5)	
Be_5Co	*C*15*b*	5.852	
YCo_2	*C*15	7.22	221
La_3Co	DO_{11}	7.277(9); 10.02(10); 6.575(8)	228
$Co_{13}La$	$D2_3$	11.344	
$CeCo_5$	*D*2*d*	4.933; 4.011	
Co_2Tb	*D*2*d*	5.090; 12.52	
Co_5Ho	*D*2*d*	4.913 8; 3.975 0	
Gd_3Co	DO_{11}	7.027; 9.496; 6.296	235
GdCo	Bf	3.90; 4.87; 4.22 ($M = 2$)	
Gd_2Co_3	cubic	7.98 ($M = 6$)	
$GdCo_2$	*C*15	7.359	
$GdCo_3$	rhomb.	5.039; 24.522; 120°	
$GdCo_4$	hex.	5.47; 6.02 ($M = 2$)	
$GdCo_5$	*D*2*d*	4.960; 3.989	
$CeCo_2$	*C*15	7.15	
$TiCo_2$	*C*36	4.73; 15.41 (Co-rich)	
	*C*15	6.692	
TiCo	*B*2	2.993	
Ti_4Co_2O	$E9_3$	11.30 (Rostoker[32])	
Ti_2Co	cubic f.	11.283	
$ZrCo_2$	*C*15	6.950	
$ThCo_3$	rhomb.	5.03; 24.54; 120°	
UCo_2	*C*15	6.978 3(4)	
UCo	*Ba*	6.352 5	
U_6Co	*D*2*c*	10.36; 5.21	
VCo_3	DO_{21}	5.037; 12.29	229
V_3Co	*A*15	4.676	
$NbCo_2$	*C*15	6.760	
Co_2Ta	*C*36	4.747; 15.47	
Co_3Ta (α)	*C*15	6.788	
(β)	*C*36	4.728; 15.45	230
Co_2Ta (α)	*C*14	4.838; 7.835	
(β)	*C*15	6.752	
(γ)	*C*36	4.747; 15.47	
CrCo (σ)	*D*8*b*	8.84(1); 4.60(1)	36
$MoCo_3$	$D0_{19}$	5.194; 4.212	
Mo_6Co_7	$D8_5$	4.761(2); 25.58(2); 120°	
Mo_3Co_2	*D*8*b*	9.279; 4.871	
WCo_3	$D0_{19}$	5.120; 4.120	
W_6Co_7	$D8_5$	4.739 8(17); 25.542(11); 120°	
FeCo	*B*2	2.850 4 } transf. temp. ~730°C; constants at room temp.	
	*A*2	2.855 0 }	
Ni	*A*1	3.535 0	
BeNi	*B*2	2.625	
$Be_{21}Ni_5$	cubic	7.619	
$MgNi_2$	*C*36	4.807; 15.77	
Mg_2Ni	*Ca*	5.205(1); 13.236(2)	
$CaNi_5$	*D*2*d*	4.952; 3.937	
YNi_2	*C*15	7.182	221
YNi_3	hex.	4.977(4); 24.44(2) }	231
YNi	ortho.	7.156(3); 4.124(1); 5.515(2) }	
YNi_5	*D*2*d*	4.867; 3.973	
$LaNi_5$	*D*2*d*	5.016; 3.983	
$LaNi_2$	*C*15	7.340	
$CeNi_5$	*D*2*d*	4.879; 3.994	
$CeNi_2$	*C*15	7.208(5)	

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Group VIII: Fe, Co, Ni; Ru, Rh, Pd; Os, Ir, Pt—*continued*			
$CeNi_3$	$D0_{21}$	4.945(5); 16.48(1)	233
$PrNi_5$	$D2d$	4.967 5; 3.972 5	
$PrNi_2$	$C15$	7.287	
Gd_3Ni	$D0_{11}$	6.95; 9.68; 6.36	234
Gd_3Ni_2	tetr.	7.28; 8.61 ($M = 4$)	
GdNi	$B27$	5.428(2); 4.353(2); 6.931(2)	
$GdNi_2$	$C15$	7.213	
Gd_2Ni_7	hex.	4.953; 24.21	
$GdNi_4$	hex.	5.35; 5.83 ($M = 2$)	
$GdNi_5$	$D2d$	4.909; 3.965	
Gd_2Ni_{17}	Th_2Ni_{17}	8.43; 8.04	
Ni_5Sm	$D2d$	4.925; 3.960	
Ni_5Tb	$D2d$	4.894(5); 3.966(5)	232
Ni_5Ho	$D2d$	4.867; 3.966	
Ni_5Yb	$D2d$	4.842; 3.959	
Ti_2Ni	O_h^7–$Fd3m$	11.324	236
$TiNi_3$	$D0_{24}$	5.102 8; 8.271 9	
TiNi	$B2$	3.007	
Ti_2Ni	$E9_3$	11.29	
Ti_4Ni_2O	$E9_3$	11.30	
$ZrNi_5$	$C15b$	6.702(4)	38
Zr_2Ni	$C16$	6.483(3); 5.267(5)	238
$ZrNi_2$	$C15$	6.915 5	
$ZrNi_3$	$D0_{19}$	5.215(8); 4.244 2(6)	
ZrNi	ortho.	3.268; 9.937; 4.101	
Hf_2Ni	$C16$	6.405; 5.252	
HfNi	Bf	3.218; 9.788; 4.117	
$ThNi_5$	$D2d$	4.941; 4.000	
$ThNi_2$	$C32$	3.960(3); 3.844(4)	
UNi_5	$C15$	6.785	
UNi_2	$C14$	4.963 0(5); 8.235(1)	
U_6Ni	$D2c$	10.37; 5.21	
$PuNi_4$	monocl. $62/m$	4.87; 8.46; 10.27; $\beta = 100°$ ($M = 6$)	239
$PuNi_3$	rhomb.	5.00; 24.35; 120°	240
PuNi	Bf	3.59; 10.21; 4.22	241
VNi_3	$D0_{22}$	3.54; 7.22	
VNi_2	orh.	2.559; 7.641; 3.549	
$V_{19}Ni_{11}$	$D8b$	8.97; 4.64	
$NbNi_3$	$D0_{22}$	3.62; 7.41	
$TaNi_3$	$D0a$	5.122; 4.235; 4.522	
Cr_2Ni	$A2$(tetr. deformed)	($c/a = 1.09$; metastable intermediate by quenching)	
$MoNi_4$	$D1a$	5.683; 3.592	
$MoNi_3$	$D0a$	5.064; 4.223; 4.449(2)	242
WNi_4	$D1a$	5.730(1); 3.553(1)	
$MnNi_3$	$L1_2$	3.593 (<510°C)	
MnNi	$L1_0$	3.69; 3.49 (<~700°C)	
	$A2$	2.974 (range ~700–900°C; constant at 745°)	
$FeNi_3$	$L1_2$	3.5550 (<586°C)	
Fe_5Cr_5Ni	cubic b.c.	8.88 (powder diagram similar $A12$-type)	
Ru	$A3$	2.705 8; 4.281 6; 2.705 3; 4.282 0	
Ru_2Be_3	$D5_3$	11.42	252
$RuBe_2$	$C14$	5.96; 9.18	
Ru_3Be_{10}	$D8_2$	11.03	
Ru_2Ce	$C15$	7.539	244
TiRu	$B2$	3.07	40
$ZrRu_2$	$C14$	5.131; 8.490	
URu_3	$L1_2$	3.977(3)	39
RuU_2	monocl.	13.106; 3.343; 5.202; 96.17°($M = 4$)	245
Rh	$A1$	3.803 0	
RhMg	$B2$	3.099(2)	246
RhBe	$B2$	2.739 7(5)	

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Group VIII: Fe, Co, Ni; Ru, Rh, Pd; Os, Ir, Pt—*continued*			
Rh_2Sr	$C15$	7.695	
Rh_3Ti	$L1_2$	3.823	248
Rh_3Hf	$L1_2$	3.911	248
Rh_3Nb	$L1_2$	3.865	
Rh_3Zr	$L1_2$	3.887(1)	
Rh_3V	$L1_2$	3.795	
Rh_3Ta	$L1_2$	3.86	
Pd	$A1$	3.887 4(1)	
BePd	$B2$	2.813	
Be_5Pd	$C15b$	5.982 (ordered as Be_4Be, Pd)	
	$C15b$	5.982 [disordered as Be_4 (Be, $Pd)_2$]	
$Be_{12}Pd$	$D2b$	7.271(5); 4.251(5)	
Pd_2Ca	$C15$	7.652	
Pd_2Sr	$C15$	7.800	247
Pd_2Ba	$C15$	7.967	
$TiPd_3$	$D0_{24}$	5.489; 8.964	
Pd_3Zr	$D0_{24}$	5.611 9(2); 9.231 6(5)	248
Pd_3Hf	$D0_{24}$	5.593 5(2); 9.192 8(5)	248
Pd_4Th	$L1_2$	4.113(3)	249
$PdTi_2$	$C11b$	3.09; 10.04	342
$PdZr_2$	$C11b$	3.299(1); 10.88(1)	342
$PdHf_2$	$C11b$	3.251; 11.061	342
PdMg	$B2$	3.17(1)	341
UPd_3	$D0_{24}$	5.770; 9.652	39
Pd_2V	ortho.	2.750; 8.250; 3.751	250
Pd_3V	$D0_{22}$	3.847; 7.753	
$FePd_3$	$L1_2$	3.848	
FePd	$L1_0$	3.855(2); 3.714(2)	
Os	$A3$	2.733 8; 4.319 5	
$Be_{12}O_5$	monocl.	10.628(6); 8.480 3(3); 11.305(3); 96.55(1)°	
TiOs	$B2$	3.081	40
$ZrOs_2$	$C14$	5.20; 8.53	
UOs_2	$C15$	7.514(4)	39
TaOs (σ)	$D8b$	9.91(6); 5.10(8)	41
OsW_2 (σ)	$D8b$	9.63; 4.98	41
Ir	$A1$	3.839 0	
Ir_2Sr	$C15$	7.849	247
Ir_3Ti	$L1_2$	3.847	250
Ir_3Hf	$L1_2$	3.935	
Ir_3Nb	$L1_2$	3.890(2)	
Ir_3Zr	$L1_2$	3.94	
Ir_3V	$L1_2$	3.812	
$ZrIr_2$	$C15$	7.36	
UIr_2	$C15$	7.495 4	39
IrW	$B19$	4.452; 2.760; 4.811	
Pt	$A1$	3.923 3	
Pt_2Ca	$C14$	5.373; 9.311	
Pt_2Sr	$C15$	7.742	
Pt_2Ba	$C15$	7.947	
PtMg	$B20$	4.863	341
Pt_3Mg	$L6_0$	3.88; 3.72	341
$CePt_2$	$C15$	7.725	
$TiPt_3$	$D0_{24}$	5.520; 9.019	
Ti_3Pt	cubic.	5.030	43
$ZrPt_3$	$D0_{24}$	5.624; 9.213	
Pt_3Hf	$D0_{24}$	5.636; 9.208	248
Pt_3U	$D0_{19}$	5.748; 4.898	39
Pt_2U	ortho.	5.60; 9.68; 4.12	251
$FePt_3$	$L1_2$	3.87(2) (<700°C)	
FePt	Aa	3.905; 3.735	

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Group VIII: Fe, Co, Ni; Ru, Rh, Pd; Os, Ir, Pt—*continued*			
Fe_3Pt	$L1_2$	3.730 (<850°C)	
$CoPt_3$	$L1_2$	3.831 (constant at 700°C; transformation to disordered state ~750°C; constant at 800°C = 3.829)	
CoPt	$L1_0$	3.80; 3.70 (constant at 700°C; transformation temp. to disordered *A*1-type ≈ 825°C)	
NiPt	$L1_0$	3.821; 3.591 (<600°C)	
Group Ib: Cu, Ag, Au			
Cu	*A*1	3.613	
BeCu (γ)	*B*2	2.702(3)	
Be_2Cu	*C*15	5.977 to 5.97	
Be_3Cu	hex.	4.179(3); 2.557(3)	
ZrIr	*B*2	3.318	
Zr_2Ir	*C*16	6.51; 5.62	
$MgCu_2$	*C*15	7.048	
Mg_2Cu	*Cb*	9.07; 5.284; 18.25	
$CaCu_5$	*D*2*d*	5.092; 4.086	
$LaCu_5$	*D*2*d*	5.186; 4.110	
CeCu	*B*27	7.19 (14); 4.30(8); 6.23(12)	253
$CeCu_2$	ortho. 1*mma*	4.433; 7.064; 7.472	254
$CeCu_6$	ortho. *Pnma*	8.109; 5.098; 10.172	255
$CeCu_5$	*D*2*d*	5.104; 4.112	
Cu_5Tb	*D*2*d*	5.030; 4.090 (also *C*15*b* 7.041)	256
Cu_5Sm	*D*2*d*	5.07; 4.104	256
Ti_3Cu	*L*1*a*	4.158; 3.594	
$CuTi_2$	*C*11*b*	2.953; 10.734	
Ti_2Cu	$D2_3$	11.24	
Ti_4Cu_2O	$E9_3$	11.47	
TiCu (δ)	$L1_0$	3.140; 2.856	
(γ)	*B*11	3.107; 5.919	
$TiCu_3$ (β)	*A*20	2.585; 4.527; 4.351 (>~600°C)	
(β)	$L6_0$	5.169; 4.342; 4.531 (<~ 600°C)	
$CuZr_2$	*C*11*B*	3.220 4; 11.183 2	342
$CuHf_2$	*C*11*B*	3.164 3; 11.153	342
$ThCu_2$	*C*32	4.383(4); 3.496(3)	47
Th_2Cu	*C*16	7.324(5); 5.816(5)	47
UCu_5	*C*15*b*	7.03	
$PdCu_3$	$L1_2$	3.722 (also A1 3.701)	
	tetr.	3.710(3); 25.655(21)	
PdCu	*A*1	3.766	
PtCu	*A*1	3.776 9(5)	
Ag	*A*1	4.086	
LiAg	*B*2	3.168	
Be_2Ag	*C*15	6.287	
MgAg	*B*2	3.311 4	
Mg_3Ag	cubic	17.622(7)	
Ca_5Ag_3	tetr.	8.039; 15.011	258
LaAg	*B*2	3.814	
CeAg	*B*2	3.785(16)	
PrAg	*B*2	3.723	
TiAg	$L1_0$	4.104; 4.077 (also B11 2.90; 8.14)	
ZrAg	*B*11	3.471; 6.603	
$AgZr_2$	*C*11*B*	3.253; 7.952	342
Th_2Ag	*C*16	7.581; 5.854	48
Pt_3Ag	$L1_2$	3.900	
Au	*A*1	4.07894(5)	
$NaAu_2$	*C*15	7.85(1)	
Na_2Au	*C*16	7.415(5); 5.522(5)	
BeAu	*B*20	4.668(1)	
Be_5Au	*C*15*b*	6.100	
MgAu	*B*2	3.266	

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Group Ib: Cu, Ag, Au—*continued*			
Mg_3Au	$D0_{18}$	4.64; 8.46	
$TiAu_4$	*D1a*	6.460(7); 3.975(8)	
$TiAu_2$	*C11b*	3.419; 8.514	
Ti_3Au	$L1_2$	4.096	
Ti_3Au	*A15*	5.088 9(6)	43
Th_2Au	*C16*	7.462; 5.989	48
V_3Au	*A15*	4.880	49
Nb_3Au	*A15*	5.205	49
MnAu	ortho.	3.146; 3.182; 3.291	
Au_2Mn	*C11b*	3.359; 8.746	259
Au_4Mn	*D1a*	6.45; 4.03	260
CuAu	$L1_0$	3.968; 3.662	
	A1	3.872	
	ortho.	3.676(2); 3.956(3); 39.72(2)	
Cu_3Au	$L1_2$	3.747	
Group IIb: Zn, Cd, Hg			
Zn	*A3*	2.664 74(4); 4.946 9(1)	
LiZn	*B32*	6.234	
$NaZn_{13}$	$D2_3$	12.283 6(3)	
KZn_{13}	$D2_3$	12.335(5)	
Mg_2Zn_{11}	*D8c*	8.552(5)	
$MgZn_2$	*C14*	5.223(1); 8.566(3)	
$CaZn_{13}$	$D2_3$	12.185(5)	
$CaZn_5$	*D2d*	5.371; 4.242	
$SrZn_5$	orthorh. *Pnma*	13.147(7); 5.312(2); 6.707(3)	50
$SrZn_5$	*D2d*	5.549; 4.283	
$SrZn_{13}$	$D2_3$	12.242(2)	
$BaZn_5$	orthorh. *Cmcm*	10.788(8); 8.441(5); 5.318(3)	50
$BaZn_{13}$	$D2_3$	12.359(5)	
$LaZn_5$	*D2d*	5.416; 4.217	
LaZn	*B2*	3.76(1)	
CeZn	*B2*	3.70(1)	
PrZn	*B2*	3.692	
$TiZn_3$	$L1_2$	3.932 2(3)	51
$TiZn_2$	*C14*	5.064; 8.210	51
Th_2Zn	*C16*	7.614(4); 5.658(4)	47, 52
$ThZn_2$	*C32*	4.20; 4.17	261
$ThZn_4$	*D13*	4.273; 10.359	261
$ThZn_9$	*D2d*	5.237; 4.442	52
$ZrZn_{22}$	f.c.c.*Fd*/3*m*	14.103(1)	262
$ZrZn_6$	tetr.	12.7; 8.68	262
$NbZn_2$	*C36*	5.05; 16.32	264
$MnZn_{13}$ (ξ)	monocl. c.	13.483(5); 7.662 6(10); 5.134(3); 127.78(4)°	
$MnZn_3$ (α)	$L1_2$	3.86 (<320°C)	
~MnZn (β)	*B2*	3.070	
(β_1)	*A3*	2.733 9; 4.455 0	
$FeZn_{13}$(ξ)	monocl.	13.424(5); 7.608 0(10); 5.061(3); 127.30(4)°	
$CoZn_{13}$(ξ)	monocl.	13.306(33); 7.535(2); 4.992(12); 126.78°	
CoZn	*A13*	6.319	
NiZn (β_1)	$L1_0$	3.895; 3.214	
(β)	*B2*	2.908 3	
Pd_2Zn	*C23*	5.35; 7.65; 4.14	341
Pd_2Zn_3 (β_1)	*B2*	3.049(>~600°C)	
PdZn (β)	$L1_0$	4.100; 3.346	53
Pd_2Zn	*B2*	3.055 (>~600°C)	
PtZn (ξ)	$L1_0$	4.05; 3.51	
Pt_3Zn	$L1_2$	3.893 (<~800°C)	
$CuZn_3$ (δ)	*A2*	3.001(8) (stable between 550–700°C)	

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Group IIb: Zn, Cd, Hg—*continued*			
Cu_5Zn_8 (γ)	$D8_2$	8.859(3)	
CuZn (β)	*A*2	(>450°C) 2.95	
(β')	*B*2	(<~450°C) 2.959	
Cu_2TiZn	*CsCl* (B2) type	2.96	265
$AgZn_3$	*A*3	2.823 1(2); 4.440 7(3)	
Ag_5Zn_8	$D8_2$	9.351(3)	54
AgZn (β')	*B*2	3.088(3)	
(ξ)	*Bb*	7.636 0; 2.819 7	
$(Ag_{0.10}Zn_{0.90})_2$ Mg	hex.	5.21; 90.3 (long period structure)	
$AuZr_2$	*C*11*B*	3.28; 11.6	342
$AuHf_2$	*C*11*B*	3.230 9; 11.605 7	342
$AuZn_3$	*cub* $Pm\bar{3}n$	7.903	266
Au_5Zn_3	ortho.	6.345; 8.971; 4.486	
AuZn	*B*2	3.14	
Au_3Zn	tetr.	5.586(5); 33.4(1)	
	ortho.	16.650; 5.585; 5.594	
Cd	*A*3	2.979; 5.617	
$LiCd_3$	*A*3	3.083; 4.889	
LiCd	*B*32	6.715(2)	
Li_3Cd	*A*1	4.250	
$RbCd_{13}$	$D2_3$	13.88	
$CsCd_{13}$	$D2_3$	13.89	
$MgCd_3$	$D0_{19}$	6.233 5; 5.044 9	
MgCd	*B*19	5.0051(3); 3.221 7(2); 5.270 0(3)	
Mg_3Cd	$D0_{19}$	6.310; 5.080	
$CaCd_2$	*C*14	5.996(2); 9.654(2)	
SrCd	*B*2	4.003	55
BaCd	*B*2	4.213(2)	55
$BaCd_{11}$	tetr.	12.040(5); 7.740(5)	56
LaCd	*B*2	3.910	
CeCd	*B*2	3.855	
PrCd	*B*2	3.827	
TiCd	*B*11	2.904; 8.954 ($M = 4$)	267
Ti_2Cd	*C*11*b*	2.865(3); 6.71(1)	267
$ZrCd_2$	cubic f.c.	4.376 8	51
ZrCd	*B*11	3.124; 8.75	51
$NaCd_2$	$Fd\bar{3}m$	30.56	268
PdCd	$L1_0$	4.281 7(1); 3.631 0(1)	53
~$PtCd_5$ (γ)	*b.c.c*	19.804 2(9) (also f.c.c. 9.9200(3))	
PtCd	$L1_0$	4.183 5; 3.816 3	53
Cd_3Cu_4	tetr.	13.701; 9.944 (also f.c.c. 25.871(2))	
$AgCd_3$	*A*3	3.071; 4.816 2	
AgCd	*A*3	2.975 0; 4.825 5	269
AgCd (β)	*A*2	(>440°C) 3.382	
(β_1)	*A*3	(230–440°C) 3.007; 4.852	
(β')	*B*2	(<210°C) 3.332	
$AuCd_3$	$D0_{21}$	8.147(2); 8.511(2)	
Au_2Cd	*A*3	2.919 8; 4.804 5	
AuCd (β_1)	*B*2	3.323 2(5) (>60–80°C)	
(β')	*B*19	4.764 5(5); 3.154 0(5); 4.864 4(5) (<~60°C)	
Au_3Cd (α_1)	$L1_0$	4.111; 4.133 (<~400°C)	
Hg	*A*10	3.458; 6.684; 120°	59
$LiHg_3$	$D0_{19}$	6.240; 4.794	
LiHg	*B*2	3.286(1)	
Li_3Hg	$D0_3$	6.548	
NaHg	ortho.	7.19; 10.79; 5.21 ($M = 8$)	60
Na_3Hg_2	tetr.	8.52; 7.80 ($M = 4$)	60
KHg	triclinic	6.59; 6.76; 7.06 ($\alpha = 106°5'$, $\beta = 101°52'$; $\gamma = 92°47'$)	61, 62

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Group IIb: Zn, Cd, Hg—*continued*			
K_5Hg_7	ortho. *Pbcm*	10.06; 19.45; 8.34 ($M = 4$)	270
KHg_2	ortho.	8.10; 5.16; 8.77 ($M = 4$)	61, 62
$MgHg_2$	$C11_b$	3.83; 8.781	
MgHg	*B*2	3.448 0(1)	
Mg_5Hg_3	$D8_8$	8.259; 5.931	
Mg_3Hg	$D0_{18}$	4.83; 8.62	
SrHg	*B*2	3.955	55
BaHg	*B*2	4.131	55
$BaHg_{11}$	*D*2*e*	9.586	
$LaHg_4$	cubic b.c.	10.968	
$LaHg_3$	$D0_{19}$	6.816; 4.971	
$LaHg_2$	*C*32	4.956; 3.637	
LaHg	*B*2	3.864	
CeHg	*B*2	3.808	
PrHg	*B*2	3.791	
TiHg	$L1_0$	4.26; 4.04	51
Ti_3Hg	*A*15	5.187(1)	51
	$L1_2$	4.165 4(4)	51
Hg_3Th	*A*3	3.361; 4.905	271
HgTh	*A*1	4.80 (high temp. phase >500°C)	
ZrHg	$L1_0$	4.45(1); 4.17(1)	51
$ZrHg_3$	$L1_2$	4.368	51
Zr_3Hg	*A*15	5.560(2)	51
$ThHg_3$	*A*3	3.361; 4.905	47
ThAg	f.c.c.	4.80	272
NdHg	*B*2	3.772	
UHg_3	$D0_{19}$	6.654(9); 4.888(5)	
UHg_2	*C*32	4.976; 3.218	
$NiHg_4$	b.c.c.	6.016	
PdHg	$L1_0$	4.271 6; 3.706 7	
Pt_3Hg	*A*1	3.979 5(5)	341
Au_3Hg	*A*3	2.918 0; 4.811 3	
CdHg	*A*6	3.932; 2.872	
$CdHg_2$	$C11_b$	3.965; 8.607	63
Group IIIb; Al, Ga, In, Tl			
Al	*A*1	4.049 50(12)	
LiAl	*B*32	6.370 3(5)	
Mg_2Al_3 (β)	*C*14	5.73; 9.54 (several other structures exist)	
$Al_{18}Mg_3Cr_2$	f.c.c.	14.53(1)	273
Y_3Al_2	tetr.	8.258; 7.702 5	
YAl	*B*33	3.870 7; 11.599; 4.396 0	274
Yal	*B*2	3.754(12)	
$CaAl_4$	$D1_3$	4.353; 11.07	
$CaAl_2$	*C*15	8.040	
SrAl	cubic	12.76	
$SrAl_4$	$D1_3$	4.460; 11.20	
$BaAl_4$	$D1_3$	4.566; 11.250	66
$LaAl_4$	$D1_3$	4.48; 10.42	
$LaAl_2$	*C*15	8.125	
CeAl	ortho.	9.27; 7.68; 5.76	67
$CeAl_4$	$D1_3$	4.380; 10.030	
$CeAl_2$	*C*15	8.060	
NdAl	ortho.	5.940; 11.728; 5.729	
$TiAl_3$	$D0_{22}$	3.853 7(3); 8.583 9(13)	
Ti_3Al (α_2)	$D0_{19}$	5.780; 4.647	60
TiAl	$L1_0$	4.001; 4.071	
$ZrAl_3$	$D0_{23}$	4.007 4(6); 17.286(4)	
Zr_3Al	$L1_2$	4.373	82
Zr_2Al	*C*7	4.882; 5.918	277

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Group IIIb; Al, Ga, In, Tl—*continued*			
Zr_2Al_3	ortho. *Fdd*2	9.617(3); 13.934(3); 5.584(2)	278
Zr_4Al_3	hex.	5.424(1); 5.405(2)	279
Zr_3Al_2	tetr.	7.633 3(8); 6.996(2)	280
HfAl	*B*33	3.240; 10.803; 4.278	281
Hf_2Al_3	ortho.	9.474; 13.737; 5.501	282
$ThAl_3$	$D0_{19}$	6.499; 4.626	48
$ThAl_2$	*C*32	4.412(13); 4.173(13)	48
Th_3Al_2	tetr. *P*4/*mbm*	8.134; 4.221	48
Th_2Al	*C*16	7.614; 5.857	48
UAl_4	*D*1*b*	4.42; 6.30	
UAl_3	$L1_2$	4.262; 13.63	
UAl_2	*C*15	7.78	
$NpAl_2$	*C*15	7.79	70
$NpAl_3$	$L1_2$	4.260(1)	70
$NpAl_4$	*D*1*b*	4.42; 6.26; 13.71	70
$PuAl_3$	hex.	6.082; 14.433	81
VAl_3	$D0_{22}$	3.780; 8.322	
$NbAl_3$	$D0_{22}$	3.841(1); 8.609(2)	
Nb_2Al	tetr.	9.945; 5.171	283
Nb_3Al	cubic	5.186	337
$TaAl_3$	$D0_{22}$	3.851 3; 8.558 3	
Cr_2Al	*C*11*b*	3.005; 8.649	
$Cr_2Mg_3Al_{18}$ (E phase)	$Fd\bar{3}m$	14.53(1)	76
$MoAl_{12}$	b.c.c.	7.573	284
$MoAl_7$	monocl.	5.12; 13.0; 13.5($\beta = 95°$)	
WAl_4	monocl.	5.272; 17.771; 5.218; 100° 12′ (a/cell = 30)	285
Mo_3Al	*A*15	4.950(1)	
WAl_5	hex. $P6_3$	4.902 0; 8.857(1)	78
WAl_{12}	cubic. b.c., $m\bar{3}$	7.580 3(5)	77
$MnAl_6$	ortho. c	7.551(4); 6.499 4(3); 8.872 4(17)	79
$MnAl_4$	ortho. p	6.795; 9.343; 13.897	
MnAl	$L1_0$	3.92; 3.54	
	tetr. p	3.92; 3.57	
Mn_3Al_{10}	hex. p	7.543; 7.898	287
Mn_4Al_{11}	tricl.	5.095(4); 8.879(8); 5.051(4); α 89.35(7)°; β 100.47(5)°; γ 105.08(6)°	286
$MnAl_3$	ortho. *Pnma*	14.79; 12.42; 12.59 ($M = 36$)	288
(Mn, Cr)Al_{12}	cubic b.c., $m\bar{3}$	7.509(1)	77
Tc_2Al_3	$D5_{13}$	4.16; 5.13	
$TcAl_{12}$	b.c.c. ($WA1_{12}$)	7.528(1)	289
$TcAl_4$	monocl.c	5.1(1); 17.0(1); 5.1(1); $\beta = 100(1)°$	
$ReAl_6$	isomorph with $MnAl_6$	7.609 1(9); 6.611 7(6); 9.023(10)	275
$ReAl_{12}$	WAl_{12}	7.5270(5)	
Fe_2Al_5	ortho. c	7.644; 6.411; 4.220	80
$FeAl_2$ (ν)	tricl.	4.88; 6.46; 8.80; $\alpha = 91.70°$; $\beta = 73.3°$; $\gamma = 96.90°$	
$FeAl_3$	monocl. c	15.509(3); 8.066(2); 12.469(2); $\beta = 107.72(2)°$	
FeAl (β_2)	*B*2	2.907 6	
Fe_3Al (B_1)	$D0_3$	5.791 (disordered A2 = 2.911)	
Co_2Al_9	*D*8*d*	10.149(6); 6.290(5); 8.556 5(5); $\beta = 142.40°$	
Co_2Al_5	$D8_{11}$	7.656 0; 7.593 2	
CoAl	*B*2	2.863	
$NiAl_3$	$D0_{20}$	6.598; 7.352; 4.802	
Ni_2Al_3	$D5_{13}$	4.036(1); 4.897(1)	
NiAl	*B*2	2.848(14)	
Ni_3Al	$L1_2$	3.571 8(2)	
Ni_2TiAl	$L2_1$	5.865	
$NiFe_2Al$ (β')	$L2_1$	5.758	
RuAl	*B*2	3.036	290
$RuAl_2$	*C*54	7.99; 4.71; 8.76	

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Group IIIb: Al, Ga, In, Tl—*continued*			
$RuAl_3$	$D0_{24}$	4.81; 7.84	
Pd_2Al_3	$D5_{13}$	4.219(1); 5.161(1)	
PdAl	hex.	15.659(4); 5.251(2)	83
	*B*2	3.0532 (560–700°C)	
OsAl	*B*2	3.001(1)	83
IrAl	*B*2	2.985	83
PrAl	ortho.	5.964(5); 11.777(10); 5.745(5) (also E1a type high-temp. phase 9.146; 7.625; 5.698)	83
$PtAl_2$	*C*1	5.910	
Pt_5Al_3	ortho.	5.413(1); 10.73(1); 3.950(1)	338
RhAl	*B*2	2.9806(5)	
Rh_2Al_5	$D8_{11}$	7.893; 7.854	
Rh_2Al_9	*D*8*d*	8.721; 6.428; 6.352; 94° 48′	
Pt_3Al	$L1_2$	3.877 5	
$CuAl_2$	*C*16	6.067(1); 4.877(1)	
CuAl (η)	monocl.	12.066; 4.105; 6.913; 55.04°	
Cu_3Al_2 (γ_2)	*C*7	4.146(1); 5.063(3)	
Cu_9Al_4 (γ)	$D8_3$	8.7068(3)	
$Mg_2Cu_6Al_5$	*D*8*c*	8.311(3)	
$MgCuAl_2$	*E*1*a*	4.00; 9.23; 7.14	
MgCuAl	*C*36	5.102; 16.76 (also other hex. structures)	
$MnCu_2Al$ (β')	$D0_3$	5.968	
$FeCu_2Al_7$	*E*9*a*	6.336(1); 14.870(2)	
Ag_2Al (γ)	*A*3	2.877 9; 4.622 5	
Ag_3Al (β')	*A*13	6.942	
(β)	*A*2	3.24(3) (hightemp. phase)	
$AuAl_2$	*C*1	5.995(1)	
Au_4Al	*A*13	6.92	
Ga	*A*11	4.524 2; 7.661 8; 4.519 5	
LiGa	*B*32	6.150(7)	
Mg_5Ga_2	$D8_9$	13.708; 7.017; 6.020	
Mg_2Ga	hex.	7.794(2); 6.893(1)	
$CaGa_2$	*C*32	4.325; 4.327	
$LaGa_2$	*C*32	4.320; 4.427	
$CeGa_2$	*C*32	4.321; 4.320	
$PrGa_2$	*C*32	4.281 7; 4.289 8	
α_2GaTi_3	$D0_{19}$	5.750(4); 4.636(3)	291
TiGa	$L1_0$	3.968; 3.970	85
Ti_2Ga	$B8_2$	4.51; 5.50	86
$TiGa_3$	$D0_{22}$	3.789; 8.734	
$ZrGa_3$	$D0_{22}$	3.963(4); 8.712(4)	
Zr_5Ga_3	$D8_8$	8.075; 5.738	291
Nb_3Ga	*A*15	5.176	337
V_3Ga	*A*15	4.819 4(5)	337
Mo_3Ga	*A*15	4.944(15)	
CoGa	*B*2	2.875	83
$NiGa_4$	$D8_2$	8.424(3)	
Ni_2Ga_3	$D5_{13}$	4.054(1); 4.882(2)	
NiGa	*B*2	2.886(1)	
Ni_2Ga	*A*6	3.75; 4.78	
RhGa	*B*2	3.006 3(8)	
PdGa	*B*20	4.965	
$PtGa_2$	*C*1	5.911	
Pt_2Ga_3	$D5_{13}$	4.22; 5.17	
PtGa	*B*20	4.910	
Pt_3Ga	$L1_2$	3.892 5 (several other structures exist)	
Pt_5Ga_3	ortho.	8.031; 7.440; 3.948	
$CuGa_2$ (θ)	tetr.	2.830; 5.839	
Cu_9Ga_4	$D8_3$	8.729 5	
	cubic	8.747(2)	
Cu_4Ga	*A*1	3.665	
Ag_3Ga	*A*3	2.886 9; 4.675 3	

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Group IIIb: Al, Ga, In, Tl—*continued*			
$AuGa_2$	*C*1	6.075 8(10)	
AuGa	*B*31	6.262(2); 3.463(2); 6.397 1(5)	
In	*A*6	3.252 2; 4.950 7	
LiIn	*B*32	6.792 0(6)	
NaIn	*B*32	7.332	
$MgIn_2$	cubic	4.60	
MgIn (β'')	$L1_0$	4.606; 4.421 (A6 phase also exists)	
Mg_2In	*C*22	8.27; 3.42	
α_2Ti_3In	$D0_{19}$	5.92; 4.78	291
Mn_3In	$D8_3$	9.453(4)	
ZrIn	*A*1	4.418	291
$ThIn_3$	$L1_2$	4.694	292
Ni_2In_3 (δ')	$D5_{13}$	4.387(3); 5.295(4)	
NiIn (δ)	*B*2	3.060 (>760°C)	
(ε)	*B*35	4.536(5); 4.434(5)	
Ni_2In	*C*7	4.188 9; 5.123 0	
Ni_3In (γ)	$D0_{19}$	5.332(2); 4.234(2)	
$PdIn_3$	$D8_3$	9.433 4(5) (also $D8_2$ 9.42)	
Pd_2In_3	$D5_{13}$	4.535 2(2); 5.511 7(2)	
PdIn	*B*2	3.249 5(2)	
Pd_3In	*A*6	4.06; 5.36 (also $L1_0$ 4.0729(2); 3.791 8(2))	
Pt_3In_7	*D*8*f*	9.435	
$PtIn_2$	*C*1	6.366	
Pt_3In	$L1_2$	3.992(1)	342
Pt_2In_3	$D5_{13}$	4.53; 5.51	
Cu_7In_3 (γ)	tricl.	10.071(5); 9.126(5); 6.724(4); $\alpha = 90.22°$; $\beta = 82.84°$; $\gamma = 106.81°$	
Cu_2MnIn	$D0_3$	6.206	
AuIn	tricl.	(4.30; 10.59; 3.56; $\alpha = 90.54°$; $\beta = 90.00°$; $\gamma = 90.17°$)	80
Au_3In	*D*0*a*	5.857 2; 4.735 2; 5.150 4	
Tl	*A*3	3.458; 5.518 (<230°C)	
	*A*2	3.871(2) (>230°C)	
LiTl	*B*2	3.438(3)	
NaTl	*B*32	7.448 0(4)	
MgTl	*B*2	3.628	
Mg_2Tl	*C*22	8.082 9(4); 3.679 6(4)	
Mg_5Tl_2	*D*8*g*	14.285(3); 7.328(2); 6.197(2)	
$CaTl_3$	$L1_2$	4.796	
CaTl	*B*2	3.852 2(6)	
SrTl	*B*2	4.038	
$LaTl_3$	$L1_2$	4.803	
LaTl	*B*2	3.936(3)	
$ThTl_3$	$L1_2$	4.751	292
PrTl	*B*2	3.895	
Pd_2Tl	*C*23	5.719; 4.228; 8.363	
Pd_3Tl	$D0_{23}$	4.117; 15.274 ($D0_{22}$ high temp. phase also exists)	342
PtTl	*B*35	5.605; 4.639	
$HgLi_2Tl$	*B*2	3.352(2)	
Group IVb: Si, Ge, Sn, Pb			
Si	*A*4	5.430 6	
Mg_2Si	*C*1	6.35	
$CaSi_2$	*C*12	3.855(5); 30.6(1); $\gamma = 120°$	
CaSi	*B*33	4.545(3); 10.728(10); 3.890(3)	102
BaSi	*B*33	5.028(5); 11.929(6); 4.131(3)	294
$LaSi_2$	*Cc*	4.318; 13.84	
$CeSi_2$	*Cc*	4.191; 13.949	
$PrSi_2$	*Cc*	4.29; 13.76 (ortho. phase also exists)	
$NdSi_2$	*Cc*	4.142(6); 13.65(2) (ortho. phase also exists)	
$SmSi_2$	*Cc*	4.08; 13.51 (ortho phase also exists)	
YSi	*Bf*	4.25; 10.50; 3.82 ($M = 4$)	295

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Group IVb: Si, Ge, Sn, Pb—*continued*			
YSi_2	ortho.	4.05; 3.95; 13.40 (several other structures exist)	
TiSi	*B*27	6.551(6); 3.633(3); 4.983(5)	296
Ti_5Si_3	$D8_8$	7.461 0(3); 5.150 8(1)	
Zr_2Si	*C*16	6.609(3); 5.298(3)	95, 96
Zr_5Si_3	*C*16	7.903; 5.581	95
$ZrSi_2$	*C*49	3.73; 14.60; 3.66	94
$HfSi_2$	*C*49	3.672(4); 14.57(1); 3.641(4)	94
$ThSi_2$	*Cc*	4.107; 14.07	
USi_2 (α)	*Cc*	3.99; 13.15 } Zachariasen[97]	
(β)	C32	3.839; 4.72 }	
USi	*B*27	7.585; 3.903; 4.663	
U_3Si_2	*D*5*a*	7.329 9(4); 3900 5	
U_3Si	*D*0*c*	6.033; 8.690	
$NpSi_2$	*Cc*	3.97(1); 13.70(3)	
$PuSi_2$	*Cc*	3.98(1); 13.58(5)	99
VSi_2	*C*40	4.58; 6.37	
V_3Si	*A*15	4.7272 (tetr. phase also exists)	
V_5Si_3	$D8_8$	7.135; 4.842	100
$NbSi_2$	*C*40	4.819(2); 6.592(2)	
Nb_5Si_3	$D8_8$	7.52; 5.24 (also $D8_m$ and $D8_1$ phases)	100
$TaSi_2$	*C*40	4.782; 6.565	
$CrSi_2$	*C*40	4.45; 6.529	
CrSi	*B*20	4.622(1)	
Cr_3Si	*A*15	4.556(4)	
$MoSi_2$	$C11_b$	3.206; 7.846	101
Mo_3Si	*A*15	4.898	
Mo_5Si_3	$D8_m$	9.642 5(5); 4.909 6(3)	100
WSi_2	$C11_b$	3.213; 7.887	101
Mn_4Si_7	tetr.	5.525(1); 17.463(3)	
MnSi	*B*20	4.558	
Mn_5Si_3	$D8_8$	6.912(5); 4.812(5)	
Mn_3Si	$D0_3$	5.720	
$ReSi_2$	$C11_b$	3.138; 7.666	
$FeSi_2$	tetr.	2.690 1; 5.134 (also ortho. phase)	
FeSi	*B*20	4.517(5)	
Fe_5Si_3	$D8_8$	6.759(5); 4.720(5)	
Fe_3Si	$D0_3$	5.644	
$CoSi_2$	*C*1	5.356(1)	
CoSi	*B*20	4.450	
Co_2Si	*C*37	4.919; 3.725; 7.104	
$NiSi_2$	*C*1	5.406	
NiSi	*B*31	5.190(1); 3.330(1); 5.628(2)	106
Ni_2Si (θ)	hex.	3.836(1); 4.948(1) (>981°C)	106
(δ)	*C*23	5.022(1); 3.741(1); 7.088(1) (<981°C)	106
Ni_3Si (β_1)	$L1_2$	3.505 (<1040°C)	
Ru_2Si_3	tetr.	5.52; 4.46 (also ortho. phase)	
RuSi (1)	*B*2	2.909(2)	
(2)	*B*20	4.707 5	
RhSi	*B*31	5.630; 3.061; 6.311	107
Rh_2Si	*C*23	5.370; 3.935; 7.360	
Rh_5Si_3	ortho.	5.317; 10.131; 3.895	
Rh_3Si_2	$B8_1$	3.949; 5.047 (also $B8_2$ with same lattice parameters reported)	293
RhSi	*B*31	5.630; 3.061; 6.311	
PdSi	*B*31	5.617(3); 3.386(2); 6.149(4)	
Pd_2Si	*C*22	6.496(5); 3.433(4)	108
Os_2Si_3	ortho.	11.157(2); 8.964(1); 5.580(1)	
OsSi	*B*2	2.960(1) (also *B*20 phase)	297
Ir_3Si	*D*0*c*	5.222; 7.954	
Ir_2Si	*C*23	5.284; 3.989; 7.615	
Ir_3Si_2	$B8_2$	3.963; 5.126 (also $D8_2$ with same lattice parameters reported)	293

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Group IVb: Si, Ge, Sn, Pb—*continued*			
$IrSi_3$	$D0_{18}$	4.350; 6.630 (several other structures exist)	
PtSi	*B*31	5.595; 3.603; 5.932	
Pt_2Si	*C*22	6.525; 3.603 (various tetr. structures also reported)	110
$BeZrSi$	$B8_2$	3.722; 7.232	8
Pd_4Al_3Si	*B*20	4.840	83
$Cu_{15}Si_4$ (ε)	$D8_6$	6.915	
Cu_5Si	tetr.	8.815; 7.903	
Cu_7Si	*A*3	2.560 5; 4.184 6	
$Cu_{16}Mg_6Si_7$	*D*8*a*	11.65(2) ($A = 116$)	111
Cu_3SiMg_2	*C*14	5.004(5); 7.872(8)	
$Al_{13}Cr_4Si_4$	cubic F	10.917(1)	112
Al_9Mn_3Si	*E*9*c* $Al_{10}Mn_3$	7.42; 7.72	
Al_3FeSi_2	tetr. Ga_5Pd	6.07; 9.50	
$AlNi_2Si$	*B*20	4.537	83
Ge	*A*4	5.657 7	
Mg_2Ge	*C*1	6.380 (*C*15 phase also reported)	
$CaGe_2$	*C*12	3.949; 30.72; 120°	114
Y_5Ge_3	$D8_8$	8.46(1); 6.36(1)	298
$CeGe_2$	*Cc*	4.26; 14.22	299
$PrGe_2$	*Cc*	4.43(5); 14.00(5) (ortho. phases also reported)	
$TiGe_2$	*C*54	8.588(8); 5.032(3); 8.862(6)	
Ti_5Ge_3	$D8_8$	7.563(2); 5.228(2)	
$ZrGe_2$	*C*49	3.810(3); 15.07(2); 3.769(2)	
Zr_5Ge_3	$D8_8$	8.029(2); 5.594(2)	115
Hf_5Ge_3	$D8_8$	7.941(1); 5.542(2)	298
$\alpha ThGe_2$	*Cc*	4.133; 14.202	
$ThGe_2$	*C*49	4.223(2); 16.911(6); 4.052(2) (other ortho structures also proposed)	301
$ThGe_3$	cubic	11.72	
V_3Ge	*A*15	4.763	
$NbGe_2$	*C*40	4.977(2); 6.809(3)	
$TaGe_2$	*C*40	4.938(2); 6.740(2)	
Ta_5Ge_3	$D8_8$	7.581; 5.15	115
Cr_3Ge	*A*15	4.625 2(8)	337
CrGe	*B*20	4.800(5)	
Cr_5Ge_3	*D*8*m*	9.427; 4.786	115
Mo_3Ge	*A*15	4.932	
Mn_5Ge_3	$D8_8$	7.197; 5.038	
$FeGe_2$	*C*16	5.908(3); 4.957(3)	
Fe_2Ge	$B8_2$	4.058; 5.038	
CoGe	monocl.	11.65; 3.807; 4.945; 101.1°	293
$CoGe_2$	ortho.	5.670(3); 5.670(3); 10.796(3) (pseudo tetragonal)	
NiGe	*B*31	5.381; 3.428; 5.811	
Ni_2Ge	$B8_1$	3.934; 5.056 (also *C*23 $B8_2$ structures)	
Ni_3Ge	$L1_2$	3.570 (also *B*32)	
GeRu	*B*20	4.846	302
Ge_3Ru_2	ortho.	11.436(5); 9.238(4); 5.716(2) (tetr. at high T)	
PdGe	*B*31	5.782; 3.481; 6.259	
Pd_2Ge	*C*22	6.712(1); 3.408(1)	108
$OsGe_2$	monocl.	8.995(6); 3.094(4); 7.685(2); 119.17°	
Ir_4Ge_5	tetr.	5.615; 18.308	293
$IrGe_4$	hex.	6.215; 7.784	
Ir_3Ge_7	*D*8f	8.735(1)	
IrGe	*B*31	5.600(1); 3.493(1); 6.290(1)	
PtGe	*B*31	5.719(2); 3.697(1); 6.084(2)	
Pt_2Ge	*C*22	6.68; 3.53	108
Pt_3Ge	monocl.	7.930; 7.767; 5.520; $\beta = 135.28°$	293
Pt_3Ge_2	ortho.	12.240; 7.549; 6.854	
Pt_2Ge_3	ortho.	16.441 1(21); 3.377 1(3); 6.201 7(7)	
$PtGe_2$	*C*18	6.185; 5.767; 2.908	

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Group IVb: Si, Ge, Sn, Pb—*continued*			
Cu_3Ge (1)	$D0_{18}$	4.169; 7.499 (>636°C)	
(2)	*D0a*	5.28; 4.22; 4.54; 2.64; 5.45; 4.19 (<636°C)	
(3)	hex.	4.17; 4.92 (570–750°C)	
Cu_5Ge	*A3*	2.612; 4.231	
Co_4GaGe_3	*B20*	4.64	83
Pd_5Al_4Ge	*B20*	4.87	83
Rh_5GaGe_4	*B20*	4.832	83
Sn grey	*A4*	6.489 2(1) (<13.2°C)	
metallic	*A5*	5.830 8; 3.181 0 (>13.2°C)	
$Na_{15}Sn_4$	$D8_6$	13.14(2)	
Mg_2Sn	*C1*	6.77 (*C15* type phase also documented)	
$CaSn_3$	$L1_2$	4.732(3)	
$CeSn_3$	$L1_2$	4.722	
$PrSn_3$	$L1_2$	4.716(3)	
Ti_5Sn_3	$D8_8$	8.04; 5.45	
Ti_3Sn	$D0_{19}$	5.921(3); 4.762(2)	
Hf_5Sn_3	$D8_8$	8.36; 5.710	304
$ThSn_3$	$L1_2$	4.719	292
USn_3	$L1_2$	4.608 9(23)	
$MnSn_2$	*C16*	6.659(3); 5.447(3)	
Mn_2Sn	*C7*	4.404; 5.529	
$FeSn_2$	*C16*	6.542; 5.326	
FeSn	*B35*	5.307; 4.445	
Fe_3Sn_2	hex.	5.344(5); 19.845(5)	
Fe_3Sn	$D0_{19}$	5.44; 4.36 (oxygen stabilised phase)	
$CoSn_2$	*C16*	6.363(3); 5.456(3)	
CoSn	*B35*	5.278; 4.258	
Co_3Sn_2	hex.	(high temp.) 4.162; 5.233	
Co_3Sn_2	hex.	16.445; 5.179	
Co_2MnSn	$D0_3$	5.985	116
Ni_3Sn_4	*D7a*	12.214(6); 4.060(2); 5.219(3); 105.0(1)°	
Ni_3Sn_2	ortho.	7.061 5(8); 5.153 0(4); 8.090 3(12)	
	hex.	4.146; 5.253	
Ni_3Sn	*A3*	(>900°C)	
	$D0_{19}$	(<900°C), 5.305; 4.254	
Ni_2MgSn	$D0_3$	6.097(4)	
Ni_2MnSn	$D0_3$	6.059	116
RhSn	*B20*	5.122	
Rh_3Sn_2	*C7*	4.331; 5.542	
$RhSn_2$ (1)	*C16*	6.410(3); 5.656(3); (>500°C)	
(2)	$D2_b$	4.487(1); 17.717(3); (<500°C)	117
	ortho.	6.319(5); 6.319(5); 11.97(1) (pseudo tetragonal)	118
$PdSn_4$	$D1_c$	6.388 8(1); 6.441 5(1); 11.446 2(2)	
$PdSn_2$ (1)	ortho.	6.478(3); 6.478(3); 12.155(3) (pseudo tetragonal)	117
(2)	tetr.	6.490(1); 24.378(4) (low temp. mod.)	118
PdSn	*B27*	6.31; 3.86; 6.12	
Pd_3Sn_2	*C7*	4.389; 5.703	
Pd_3Sn	$L1_2$	3.976(1) (Disorders to *A1* 3.970)	
IrSn	$B8_1$	3.98; 5.63	
$PtSn_4$	$D1_c$	6.382 3(1); 6.419 0(1); 11.366 6(2)	
$PtSn_2$	*C1*	6.433 1	
Pt_2Sn_3	$D5_b$	4.334 5; 12.960 6	
PtSn	$B8_1$	4.104(2); 5.436(2)	
Pt_3Sn	$L1_2$	4.000(1)	
Cu_6Sn_5 (η)	$B8_1$	4.192(2); 5.037(2)	119
Cu_3Sn (ε)	$D0_a$	5.49; 4.32; 4.74	119
	ortho.	5.529(8); 47.75(6); 4.323(5)	120
Cu_3Sn	$D0_a$	5.49; 4.32; 4.74	121
Cu_3Sn	$C15_6$	6.1176(10) (high temp. phase)	122
$Cu_{10}Sn_3$	hex.	7.330(4); 7.864(5)	122
$Cu_{41}Sn_{11}$	cubic	17.980(7)	
$Cu_{17}Sn_3$ (β)	*A2*	3.0261 (stable >600°C)	
Cu_2MnSn	$D0_3$	6.166	

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Group IVb: Si, Ge, Sn, Pb—*continued*			
Ag_3Sn	*D0a*	5.968(9); 4.780 2(4); 5.184 3(9)	
$AuSn_4$	*D1c*	6.512 4(1); 6.516 2(1); 11.706 5(1)	
$AuSn_2$	ortho.	6.909; 7.037; 11.789	80
AuSn	$B8_1$	4.322 2; 5.522 2	
$InSn_4$	*Af*	3.205; 2.995	63
In_3Sn	*A6*	3.456; 4.391	
Pb	*A1*	4.950 8(4)	
LiPb	*B2*	3.561	
Li_3Pb	$D0_3$	6.687(3)	124
Li_7Pb_2	hex.	4.751(2); 8.589(4)	124
Li_8Pb_3	monocl. C2/*m*	8.240; 4.757; 11.03; $\beta = 104°\ 25'$ ($M = 2$)	125
$Li_{10}Pb_3$	$D8_3$	10.082	
$Li_{22}Pb_5$	f.c.c.	20.08	305
NaPb	tetr.	10.580(5); 17.746(15)	126
$Na_{15}Pb_4$	$D8_6$	13.02	
KPb_2	*C14*	6.66; 10.76	127
Mg_2Pb	*C1*	6.836	
$CaPb_3$	$L1_2$	4.910	
$SrPb_3$	$L1_0$	4.965; 5.024	
$LaPb_3$	$L1_2$	4.902(2)	
$CePb_3$	$L1_2$	4.876(2)	
$PrPb_3$	$L1_2$	4.859 6(3)	
Ti_4Pb	$D0_{19}$	5.962; 4.814	128
UPb_3	$L1_2$	4.791	
$ThPb_3$	$L1_2$	4.853	292
$RhPb_2$	*C16*	6.674(3); 5.831(3)	
$PdPb_2$	*C16*	6.865(3); 5.844(3)	
Pd_3Pb_2	*C7*	4.465 0; 5.709 0	
Pd_3Pb	$L1_2$	4.035(1)	
IrPb	$B8_1$	3.993(5); 5.566(5)	
$PtPb_4$	$D1_d$	6.666(10); 5.978(10)	
PtPb	$B8_1$	4.26; 5.47	
Pt_3Pb	$L1_2$	4.058(1)	
Ag_4Pb	*A3*	2.92; 4.76	129
$AuPb_2$	*C16*	7.338(4); 5.658(4)	
Au_2Pb	*C15*	7.911(2)	
Group Vb: As, Sb, Bi			
As	*A7*	3.7597(1); 10.4412(2); 120°	
Li_3As	$D0_{18}$	4.450; 7.880	
LiAs	monocl. $P2_1/C$	5.79(1); 5.24(1); 10.70(2); $\beta = 117.4°$	306
Na_3As	$D0_{18}$	5.088; 8.982	
K_3As	$D0_{18}$	5.782; 10.222	
Mg_3As_2	$D5_3$	12.355	
MgLiAs	$C1_b$	6.19	
CeAs	*B1*	6.0815	
NdAs	*B1*	5.971	
ZrAs	*B1*	5.4335(6)	307
TiAs	$B8_1$	3.645; 6.109 (high temp. phase)	130
	B_i	3.64; 12.050	130
CrAs	*B31*	5.591; 3.573; 6.128 ($B8_1$ at high temp.)	
Cr_2As	*C38*	3.618(1); 6.350(1)	
~MnAs (1)	$B8_1$	3.722; 5.702 (up to 27°C)	131
(2)	*B31*	5.72; 3.676; 6.379 (40 to 125°C)	132
Mn_2As	*C38*	3.785(5); 6.265(5)	
Mn_3As	*D0d*	3.788; 3.788; 16.29 (pseudo tetr.)	
$FeAs_2$	*C18*	5.300 1(5); 5.983 8(5); 2.882 1(4)	
FeAs	*B31*	5.440 1(5); 3.371 2(4); 6.025 9(5)	
Fe_2As	*C38*	3.63; 5.98	
$CoAs_3$	$D0_2$	8.208	
$CoAs_2$	*C18*	5.051(2); 5.873(2); 3.127(1) (monocl. phase also reported)	

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Group Vb: As, Sb, Bi—*continued*			
CoAs	*B*31	5.263 1(3); 3.456 9(1); 5.801(1)	
$NiAs_2$	*C*18	4.782; 5.83; 3.545 (Rammelsbergite)	
	ortho.	5.771 8(5); 5.834 2(6); 11.421 4(12)	
NiAs	$B8_1$	3.619; 5.044	
$Ni_{11}As_8$	tetr.	6.852(7); 21.81(3)	
Ni_5As_2	hex.	6.815(1); 12.506(2)	
$PdAs_2$	*C*2	5.984 5(10)	
$PtAs_2$	*C*2	5.968 1(1)	
Cu_3As	cubic	9.619(1) ($M = 16$) (Domeykite)	
	hex.	7.102(20); 7.246(20)	
CuMgAs	*C*38	3.953; 6.225	
AgMgAs	$C1_b$	6.21	
$ZnAs_2$	monocl.	9.287(1); 7.691(2); 8.010(1); 102.47°	133
Zn_3As_2	hex.	7.27; 12.08	
ZnLiAs	*C*1	5.940(1)	
ZnNaAs	*C*38	4.176; 7.088	134
ZnAgAs	*C*1	5.912	134
Cd_3As_2	$D5_9$	8.96; 12.66	
AlAs	*B*3	5.661 0	
$AlLi_3As_2$	ortho.	11.87; 11.98; 12.11 [orthorh., deformed (D_{2h}^{27}) superstructure]	
$GeAs_2$	ortho. *Pbam*	14.76; 10.16; 3.728 ($M = 8$)	308
GaAs	*B*3	5.654	309
InAs	*B*3	6.062	309
SnAs	*B*1	5.716	
Sb	*A*7	4.300 7; 11.222; 120°	
Li_3Sb (α)	$D0_{18}$	4.701; 8.309	
(β)	$D0_3$	6.559	
NaSb	monocl.	6.80(2); 6.34(2); 12.48(4); 117.6(2)°	306
Na_3Sb	$D0_{18}$	5.355; 9.496	
K_3Sb	$D0_{18}$	6.025; 10.693 (also *C*16 phase reported)	
Cs_3Sb	*B*32	9.148 4 ($D0_3$ phase also reported)	135
Mg_3Sb_2	$D5_2$	4.573; 7.229	
CeSb	*B*1	6.388 (tetragonal phase at 9.5 Kelvin also reported)	
PrSb	*B*1	6.38	
NdSb	*B*1	6.321	
$TiSb_2$	*C*16	6.654(3); 5.806(3)	
TiSb	$B8_1$	4.115; 6.264	
Ti_3Sb	*A*15	5.222 8	310
	*D*8*m*	10.465; 5.263 9($M = 32$)	
ThSb	*B*1	6.318(1)	136
Th_3Sb_4	$D7_3$	9.3840(5)	136
$ThSb_2$	*C*38	4.34; 9.15	136
VSb_2	*C*16	6.554 1(4); 5.638 5(5)	131
V_3Sb	*A*15	4.939(1) ($M = 2$)	337
Nb_3Sb	*A*15	5.264 3($M = 2$)	337
$CrSb_2$	*C*18	6.018 3(4); 6.873 6(2); 3.270 4(2)	
CrSb	$B8_1$	4.115; 5.493	
MnSb	$B8_1$	4.130; 5.786	
Mn_2Sb	*C*38	4.080; 6.550	137
$FeSb_2$	*C*18	5.85; 6.575; 3.240	
FeSb	$B8_1$	4.124; 5.173	
$CoSb_2$	*C*18	5.600; 6.380; 3.370 (monocl. phase also reported)	
CoSb	$B8_1$	3.875 7; 5.180 2	
CoMnSb	*C*1	5.899 (other structures also reported)	
$NiSb_2$	*C*18	5.215; 6.358; 3.855	
NiSb	$B8_1$	3.932 5; 5.135 1	
Ni_3Sb	*D*0*a*	5.320 7(5); 4.280 8(3); 4.514 7(4)	
NiMgSb	$C1_b$	6.045	
Ni_2MgSb	$D0_3$	6.050(4)	
NiMnSb	$C1_b$	5.929	
Ni_2MnSb	$D0_3$	6.027	

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Group Vb: As, Sb, Bi—*continued*			
RhSb	*B*31	5.971 8(7); 3.862 1(7); 6.324(9)	
$PdSb_2$	*C*2	6.460(1)	
PdSb	$B8_1$	4.078; 5.592	
$IrSb_2$	$E1_b$	5.63; 6.573; 3.575; 90.8°	80
$PtSb_2$	*C*2	6.440 0(4)	13
PtSb	$B8_1$	4.126; 5.481	
Cu_2Sb (γ)	*C*38	4.001 4; 6.104 4	
Cu_3Sb	$D0_3$	(>400°C) 6.00 ($D0_a$ structure also reported)	
CuMgSb	$C1_b$	6.152	
CuMnSb	$C1_b$	6.100	
Ag_3Sb	$D0_a$	5.99; 4.85; 5.24 (other structures also reported)	
$AuSb_2$	*C*2	6.658 3(5)	
ZnSb	*Be*	6.23; 7.76; 8.14	
$ZnMg_2Sb_2$	$D5_2$	4.428(4); 7.185(7)	
CdSb	*Be*	6.469(1); 8.251(2); 8.522(2)	
Cd_3Sb_2	ortho.	8.40; 7.89; 12.00	
CdCuSb	*C*16	6.262 5	
AlSb	*B*3	6.135 0	309
GaSb	*B*3	6.095 1	309
InSb	*B*3	6.47	309
Tl_7Sb_2	$L2_2$	11.618(6)	
SnSb	*B*1	5.880(4)	
Bi	*A*7	4.533 0(5); 11.797(1); 120°	
LiBi	$L1_0$	4.753; 4.247	
Li_3Bi	$D0_3$	6.708	
NaBi	$L1_0$	4.89; 4.80	
Na_3Bi	$D0_{18}$	5.448; 9.655	
KBi_2	*C*15	9.501(5)	
K_3Bi	$D0_{18}$	6.178; 10.933	
Mg_3Bi_2	$D5_2$	4.666; 7.401	
MgLiBi	$C1_b$	6.74	
LaBi	*B*1	6.61	
CeBi	*B*1	6.505(2)	
PrBi	*B*1	6.47	
$ZrBi_2$	ortho.	15.49; 10.18; 3.97	312
Th_3Bi_4	$D7_3$	9.562 ($M = 4$)	311
$ThBi_2$	*C*38	4.495; 9.308	141
UBi	*B*1	6.356	141
MnBi	$B8_1$	4.28; 6.11 (ortho. phase also reported)	
NiBi	$B8_1$	4.07; 5.33	
NiMgBi	$C1_b$	6.166	
RhBi	$B8_1$	4.089 4; 5.664 2	220
Pd_5Bi_3	$B8_1$	4.48(5); 5.86(5)	
$PtBi_2$	*C*2	6.683 (several other structures also reported)	80
CuMgBi	$C1_b$	6.256	
Au_2Bi	*C*15	7.942(2)	
InBi	*B*10	5.015; 4.771	142
In_2Bi	*C*7	5.496; 6.579	
Pb_3Bi	*A*1	4.9714(3)	
Metalloids, *etc.*: B, C, P, N, Po, Te, Se, S			
B	Ag	8.75(2); 5.075(1)	
	hex.	10.96; 23.89 (105 atoms per cell)	144
	hex.	10.926 5(4); 23.809 6(13) (111 atoms per cell)	
	hex.	4.91; 12.57 (12 atoms per cell)	145
	hex.	10.932(2); 23.819(5) (141 atoms per cell)	
	tetr.	10.14(1); 14.17(1) (196 atoms per cell)	
Be_2B	*C*1	4.663	314
BeB_6	tetr.	10.16; 14.28	
MgB_2	*C*32	3.085; 3.523	146
CaB_6	cubic	4.151 2	

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Metalloids, *etc.*: B, C, P, N, Po, Te, Se, S–*continued*			
SrB_6	cubic	4.199	
BaB_6	cubic	4.252(2)	
YB_6	cubic	4.102	
LaB_6	cubic	4.156 6	
EuB_6	cubic	4.19(1)	315
TbB_6	cubic	4.102	316
TbB_4	*D1e*	7.120(3); 4.042(2)	
CeB_6	cubic	4.139 6(4)	
CeB_4	*D1e*	7.198; 4.085	
PrB_6	cubic	4.148	
NdB_6	cubic	4.128(1)	
GdB_6	cubic	4.111 1(2)	
ErB_6	cubic	4.100	
YbB_6	cubic	4.102	
TiB_2	*C*32	3.03; 3.22	
TiB	*B*27	6.105; 3.048; 4.551 (B1 structure also reported)	149
ZrB_{12}	*D2f*	7.388(3)	151
ZrB_2	*C*32	3.169 4(2); 3.530 7(4)	151
ZrB	*B*1	4.65	148 (149, 150)
ThB_6	cubic	4.100	
ThB_4	*D1e*	7.122 6(5); 4.024 2(3)	
UB_{12}	*D2f*	7.475	
UB_4	*D1e*	7.077; 3.983	
PuB	*B*1	4.905(4)	317
PuB_2	*C*32	3.186 2(2); 3.949(2)	
PuB_4	*D1e*	7.101 8(3); 4.002 8(1)	
PuB_6	cubic	4.113 4(3)	
VB_2	*C*32	2.998; 3.055	
V_3B_4	$D7_b$	3.052; 13.35; 2.98	152
VB	B_f	3.060(3); 8.048(3); 2.972(1)	
NbB_2	*C*32	3.110; 3.270	
Nb_3B_4	$D7_b$	3.342; 14.12; 3.143	
NbB	B_f	3.298; 8.722; 3.164	
TaB_2	*C*32	3.098; 3.225	
Ta_3B_4	$D7_b$	3.036; 12.846; 2.958	
TaB	B_f	3.280(1); 8.670(3); 3.155(2)	
Ta_2B	*C*16	5.783(3); 4.866(3)	
CrB_2	*C*32	2.969; 3.066	
Cr_3B_4	$D7_b$	3.000 4(8); 13.018(3); 2.951 6(8)	318
CrB	B_f (also B33)	2.978 2(7); 7.870(1); 2.934 6(7)	
Cr_5B_3	$D8_1$	5.480; 10.01	153
Cr_2B	$D1_f$	14.56; 7.320; 4.208	153
Mo_2B_5	$D8_i$	3.007, 20.91, 120°	
MoB_2	*C*32	3.039; 3.055 (*C*12 structure also reported)	
MoB	*Bg*	3.103; 16.95	
Mo_2B	*C*16	5.547(3); 4.739(3)	
W_2B_5	$D8_i$	3.011(3); 20.93(1)	
WB_2	*C*32	3.020(2); 3.050(2)	
WB (1)	B_f	3.19; 8.40; 3.07 (>1 850°C)	
(2)	B_g	3.0973(2); 16.956 7(25)	
W_2B	*C*16	5.568(3); 4.744(3)	
Mn_3B_4	$D7_b$	3.035; 12.838; 2.960	
MnB	*B*27	5.562; 2.976; 4.146	
Mn_2B	*C*16	5.148 4(6); 4.208 1(6)	
Mn_4B	$D1_f$	14.53; 7.293; 4.209	
ReB_2	hex.	2.900(1); 7.478(2)	319
FeB	*B*27	5.502; 2.947; 4.058	
Fe_2B	*C*16	5.110(3); 4.249(3)	
CoB	*B*27	5.260; 3.042; 3.954	
Co_2B	*C*16	5.012; 4.220	

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Metalloids, *etc.*: B, C, P, N, Po, Te, Se, S—*continued*			
Co_3B	$D0_{11}$	5.232; 6.636; 4.418	154
Ni_2B	$C16$	4.991(3); 4.247(3)	
Ni_3B	$D0_{11}$	5.215 73(31); 6.618 41(46); 4.391 59(21)	154
RuB	cubic	7.02	320
RuB_2	orthorh.	2.865; 4.045; 4.645	
Rh_2B	orthorh.	5.42(1); 3.98(2); 7.44(3)	155
Pd_3B_2	hex.	6.48; 3.42	
Pd_5B_2	monocl. $C2/c$	12.786; 4.955; 5.472 ($\beta = 97°\ 2'$; $M = 4$)	321
Pd_3B	$D0_{11}$	5.463; 7.567; 4.852	
OsB	cubic	7.03	
Pt_3B	tetr.	2.63; 3.83	
AlB_2	$C32$	3.005; 3.257	
AlB_{10}	orthorh.	8.881; 9.100; 5.690 ($M = 2$)	339
AlB_{12}	ortho.	16.623(5); 17.540(5); 10.180(5) (384 atoms per unit cell)	
		16.573(4); 17.510(3); 10.144(1) (396 atoms per unit cell)	
	tetr.	10.161(7); 14.283(8)	
$Al_6Mo_7B_7$	ortho.	7.03; 6.34; 5.76	
B_3Si	hex. $R\bar{3}m$	6.319; 12.713 (tetr. structure also reported)	322
C	$A4$	3.566 986(3) (diamond)	
	$A9$	2.464(2); 6.711(4)	
RbC_8	ortho.	4.926(4); 8.532(6); 22.472(10)	
CsC_8	hex.	4.945; 17.76	
Be_2C	$C1$	4.340(2)	
MgC_2	tetr.	3.92(2); 5.02(2)	
Mg_2C_3	hex.	7.43(3); 10.59(5)	
CaC_2	$C11_a$	3.89(1); 6.38(1)	
SrC_2	$C11_a$	4.11; 6.68	
BaC_2	$C11_a$	4.40; 7.06	
LaC_2	$C11_a$	4.000; 6.58	323
La_2C_3	$D5c$	8.816 2(5)	
CeC_2	$C11_a$	3.881(2); 6.487(3)	
PrC_2	$C11_a$	3.847(1); 6.428(2)	
NdC_2	$C11_a$	3.822; 6.386	
TiC	$B1$	4.327(1)	
ZrC	$B1$	4.694 1	
HfC	$B1$	4.628	
ThC_2	$C11_a$	4.14; 5.28	156
	Cg	6.684(2); 4.220(1); 6.735(2); 103.91°	157
ThC	$B1$	5.322 1(5)	
UC_2	$C11_a$	3.522(1); 5.988(1)	161
U_2C_3	$D5c$	8.074	162
UC	$B1$	4.960 6(5)	
Pu_2C_3	$D5c$	8.132	
PuC	$B1$	4.972 5	
VC	$B1$	4.172(3)	
V_4C_3	hex.	2.919(2); 27.79(3)	
NbC	$B1$	4.470	163
Nb_2C	$B8_1$	5.416 9(2); 4.971 9(6)	163
TaC	$B1$	4.4562(2)	
Ta_2C	$B8_1$	3.105; 4.940	
C_3Ta_4	hex.	3.116(5); 30.00(5)	164
Cr_7C_3	ortho.	4.526(5); 7.010(5); 12.142(5)	
Cr_7C_3	hex.	14.01; 4.53	
Cr_3C_2	$D5_{10}$	5.532 9(5); 2.829 0(2); 11.471 9(7)	
$Cr_{23}C_6$	$D8_4$	10.63	
MoC	B_h	2.903; 2.820 (several other hex. structures proposed).	
MoC	$B1$	4.26	
Mo_2C	$B8_1$	2.994 9(5); 4.373 24(6)	
WC	B_h	2.906; 2.837	
W_2C	$B8_1$	2.992; 4.725 (other hex. structures reported)	

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Metalloids, *etc.*: B, C, P, N, Po, Te, Se, S—*continued*			
Mn_7C_3	ortho.	4.545; 6.962; 11.984 (isom. with Cr_7C_3)	
$Mn_{23}C_6$	$D8_4$	10.598	
$Fe_{20}C_9$	ortho.	9.06(3); 15.69(3); 7.94(1)	165
Fe_2C	ortho.	4.704(16); 4.318(5); 2.830(6)	166
Fe_3C	$D0_{11}$	5.078 7; 6.729 7; 4.514 4 (Cementite)	
('ε')	hex.	4.767; 4.354	167
Fe (+0.25%C)	$L'2$ (deformed tetr.)	2.842; 3.008 } Martensite at low temp.	
(+0.75%C)	$L'2$ (deformed tetr.)	2.850; 2.939 }	
Fe (+C)	$A1$	3.654 4 (Austenite at high temp.)	
	$A2$	2.935 (Ferrite)	
Fe_3Mo_3C	$E9_3$	11.10	
$Fe_{21}Mo_2C_6$	$D8_4$ ordered	10.546	
Fe_3W_3C	$E9_3$	11.10	
$Fe_{21}W_2C_6$	$D8_4$ ordered	10.533	
Co_2C	$C18$	2.910; 4.469; 4.426	326
Co_3C	$D0_{11}$	5.07; 6.70; 4.53	
Co_3W_3C	$E9_3$	11.08	
$Co_3W_9C_4$	hex.	7.826; 7.826	168
$Co_3W_{10}C_4$	hex.	7.848; 7.848 (isom. Fe, Ni-compounds)	
Ni_3C	hex.	4.553; 12.92	169
CNi_3W_3	$E9_3$	11.15	170
Ni_5W_6C (η)	cubic	10.873	170
$Ni_3W_{16}C_6$	hex.	7.818 3; 7.818 0	170
Al_4C_3	$D7_1$	3.338; 25.117	
$AlMn_3C$	$E2_1$	3.872 2(1)	
~$AlFe_3C$	$E2_1$	3.78	
SiC	$B3$	4.358 1(5) } there are several other	
	$B5$	3.076; 5.048 } hexagonal modifications	
B_4C	$D1g$	5.60; 12.12	
P (red)	cubic	11.31 ($A = 66$) (also claimed to be 7.331)	
(white)	$I432$	18.51 ($A = 224$; other possible space groups; I m3 m and I43 m)	
Li_3P	$D0_{18}$	4.273; 7.594	
Na_3P	$D0_{18}$	4.980; 8.797	
Be_3P_2	$D5_3$	10.15(3)	
Mg_3P_2	$D5_3$	12.01(1)	
MgLiP	$C1$	6.023 ($C1_b$ structure also claimed)	
AlP	$B3$	5.500	
GaP	$B3$	5.449 9(7)	309
InP	$B1$	5.423	309
LaP	$B1$	6.034 6(9)	
CeP	$B1$	5.927	
PrP	$B1$	5.903	
NdP	$B1$	5.682	
Th_3P_4	$D7_3$	8.618	
U_3P_4	$D7_3$	8.203 9	
UP	$B1$	5.584 4(5)	
V_3P	tetr.	9.400; 4.750	
CrP	$B31$	5.346(6); 3.107(4); 5.999(6)	
Cr_3P	$D0_e$	9.188 7(1); 4.559 3(1)	
WP	$B31$	5.731(4); 3.248(2); 6.227(4)	
MnP	$B31$	5.240 9(3); 3.180 6(2); 5.902 7(4)	
Mn_2P	$C22$	6.059; 3.440	
Mn_3P	$D0_e$	9.178(3); 4.608(3) (isom. Cr_3P)	
FeP_2	$C18$	4.972 9(7); 5.656 8(8); 2.723 0(4)	
FeP	$B31$	5.208; 3.16; 5.812	
Fe_2P	$C22$	5.864 4; 3.456 0	
Fe_3P	$D0_e$	9.110; 4.458 (isom. Cr_3P)	
CoP	$B31$	5.06; 3.27; 5.58	
Co_2P	$C23$	5.63; 3.52; 6.60	

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Metalloids, *etc.*: B, C, P, N, Po, Te, Se, S—*continued*			
Ni_2P	$C22$	5.88; 3.36	
Ni_3P	$D0_e$	8.812; 4.373 (isom. Cr_3P)	
Rh_2P	$C1$	5.502 1(3)	
Ir_2P	$C1$	5.543	
PtP_2	$C2$	5.695 0(5)	
Cu_3P	$D0_{21}$	6.88; 7.18	
ZnP_2	tetr.	5.073 6; 18.561 (several other structures claimed)	
Zn_3P_2	$D5_9$	8.088 9; 11.406 9	
ZnLiP	$C1$	5.779	
CdP_2	tetr.	5.285 2(7); 19.787(3) (isom. with ZnP_2)	
Cd_3P_2	$D5_9$	8.753 7; 12.266 9	
AlP	$B3$	5.500	
$AlLi_3P_2$	ortho.	11.47; 11.61; 11.73	
GaP	$B3$	5.449 9(7)	
InP	$B3$	5.847	
Li_3N	hex.	3.648(1); 3.875(1) (several other hex. structures claimed)	
Be_3N_2	$D5_3$	8.134(10)	
Mg_3N_2	$D5_3$	9.96(1)	
MgLiN	$C1$	4.970(2)	
Ca_3N_2	$D5_3$	11.473(1)	
(β)	$D5_2$	3.553; 5.812 (ortho. phase also reported)	
ScN	$B1$	4.505(7)	
LaN	$B1$	5.305(2)	
CeN	$B1$	5.044	
PrN	$B1$	5.163	
NdN	$B1$	5.132(2)	
EuN	$B1$	5.017(3)	172
GdN	$B1$	4.988 0	
SmN	$B1$	5.046(2)	172
YbN	$B1$	4.786(2)	172
TiN	$B1$	4.239(1)	
$TiLi_5N_3$	$E9_d$	9.700(2)	173
ZrN	$B1$	4.585(2)	
Th_2N_3	$D5_2$	3.883(2); 6.187(4)	
UN_2	$C1$	5.299	
U_2N_3	$D5_2$	3.700(2); 5.825(3) (also 10.68, $D5_3$)	
UN	$B1$	4.887(3)	
NpN	$B1$	4.898 7(5)	
PuN	$B1$	4.904 6	
VN	$B1$	4.134 70(2)	
NbN (1)	$B1$	4.377(1)	
(2)	$B8_1$	2.967; 5.538	
(3)	B_i	2.959; 11.271	
Nb_2N	B_i	3.058(1); 4.961(1)	
TaN	B_h	2.936; 2.885	174
Ta_2N	$B8_1$	3.048; 4.918	
CrN	$B1$	4.148	
Cr_2N	hex.	4.796; 4.470	
MoN	hex.	5.725; 5.608	
Mo_2N	tetr.	4.200; 8.010	
W_2N (β)	hex.	2.89; 22.85	
Mn_3N_2	tetr.	2.974(1); 12.126(4)	
Mn_4N	$E2_1$	3.871	
Fe_2N	hex.	4.787; 4.418	
Fe_3N (ε)	$B8_1$	2.705; 4.376	
Fe_4N (ε)	$E2_1$	3.896(2)	
Fe_4N (γ')	$B1$	3.790(1)	
Fe_8N (α'')	tetr.	5.720; 6.292	
Co_2N	$C18$	4.605 6(10); 4.344 3(10); 2.853 5(5)	
Co_3N	hex.	2.66; 4.35	
Ni_3N	hex.	4.616(1); 4.298(3)	

(continued)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Metalloids, *etc.*: B, C, P, N, Po, Te, Se, S—*continued*			
Cu_3N	$D0_9$	3.813	
Zn_3N_2	$D5_3$	9.743(5)	
ZnLiN	$C1_b$	4.877(2) (ordered)	
Cd_3N_2	$D5_3$	10.79(2)	
AlN	$B4$	3.12; 4.97	
$AlLi_3N_2$	$E9d$	9.461(3)	
GaN	$B4$	3.190(1); 5.189(1)	
$GaLi_3N_2$	$E9d$	9.594(3)	
InN	$B4$	3.533(4); 5.693(4)	
α Si_3N_4	hex. D_d^2–P31c	7.818(3); 5.591(4)	327
β Si_3N_4	hex. C_{6h}^2–P6/3m	7.595(1); 2.902 3(6)	
$SiLi_5N_3$	$E9d$	9.436(2)	
Ge_3N_4	hex. $P3/c$	8.196 0(8); 5.930 1(5) (also P6/3m structure reported)	173
$GeLi_5N_3$	$E9d$	9.614(2)	173
BN	$B3$	3.615 3(4) (several other structures also reported)	
Al_5C_3N	$E9_4$	3.285; 21.65	
Po (α)	Ah	3.359(1)	
(β)	$A10$	5.093(2); 4.927(2); 120°	
PbPo	$B1$	6.590(3)	
Te	$A8$	4.312(1); 5.957(2)	175
Li_2Te	$C1$	6.504	
Na_2Te	$C1$	7.294(1)	
K_2Te	$C1$	8.148	
BeTe	$B3$	5.626 9(5)	
MgTe	$B4$	4.53(1); 7.38(2)	
CaTe	$B1$	6.356(1)	
SrTe	$B1$	6.659(6)	
BaTe	$B1$	7.005(10)	
EuTe	$B1$	6.594	
YbTe	$B1$	6.361(1)	
$TiTe_2$	$C6$	3.777(3); 6.498(6)	
TiTe	$B8_1$	3.834; 6.390	
Ti_5Te_4	tetr.	10.164; 3.772 0	328
UTe	$B1$	6.155	176
U_3Te_4	$D7_3$	9.31	176
UTe_2	$C38$	4.243(5); 8.946(5)	
VTe	$B8_1$	3.983; 6.133	
CrTe	$B8_1$	4.004 7(10); 6.241 8(13)	
$MoTe_2$	$C7$	3.510; 14.00	329
WTe_2	ortho.	3.496; 6.282; 14.073 (C7 phase also reported)	177
$MnTe_2$	$C2$	6.953 2(12)	
MnTe	$B8_1$	4.149 7; 6.76 (B1 phase also reported)	
$FeTe_2$	$C18$	5.260; 6.268; 3.876 (several other structures also reported)	
FeTe	$C38$	3.829(2); 6.288(4) } C2 structure also reported	
$CoTe_2$	$C6$	3.804; 5.405 } C2 structure also reported	
	$C18$	5.328; 6.310; 3.875 } C2 structure also reported	
CoTe	$B8_1$	3.890; 5.373	
$NiTe_2$	$C6$	3.854 2; 5.260 4	
NiTe	$B8_1$	3.965; 5.358	
$RuTe_2$	$C2$	6.391 0(2)	
$PdTe_2$	$C6$	4.036 5; 5.126 2	
PdTe	$B8_1$	4.20; 5.79	
$OsTe_2$	$C2$	6.425	
$PtTe_2$	$C6$	4.025 9; 5.220 9	
Cu_2Te	Ch	4.188; 7.251	
Cu_2Te	cubic f.c.	6.03	80
CuTe	ortho.	3.10; 4.02; 6.86	80
Ag_2Te	cubic	6.6 (high temp. mod.)	
	monocl.	5.09; 4.48; 6.96; 123.4°	
ZnTe	$B3$	6.079(5) (B1 structure also reported)	
CdTe	$B3$	6.482 (B4 structure also reported)	
HgTe	$B3$	6.453; 6.429 (Coloradoite)	175

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Metalloids, *etc.*: B, C, P, N, Po, Te, Se, S—*continued*			
Ga_2Te_3	*B*3	5.892(3)	
InTe	*B*37	8.444; 7.136	63,80
In_2Te_3	*B*3	6.135	
TeTl	b.c.tetr.	12.95(1); 6.18(1)	
Te_3Tl_2	monocl.	17.413; 6.552; 7.910; 133.16°	331
SnTe	*B*1	6.315(1)	
PbTe	*B*1	6.438 4(4) (Altaite)	
As_2Te_3	monocl.	14.357 3(9); 4.019 9(2); 9.899 0(7); 95.107(5)°	178
Sb_2Te_3	*C*33	4.264(1); 30.458(7)	179
Te_2SbTl	$F5_1$	4.439; 23.20; 120°	332
Te_2BiTl	$F5_1$	4.509; 23.13; 120°	
Bi_2Te_3	*C*33	4.385 0(2); 30.487(2); 120°	
Se (γ)	*A*8	3.910(1); 5.080(1)	
(α)	*Ak*	15.018(1); 14.713(1); 8.789(1); 93.61°	
(β)	*A*1	12.85; 8.07; 9.31; 93° 8′	
Li_2Se	*C*1	6.001 4(4)	
Na_2Se	*C*1	6.799	
K_2Se	*C*1	7.675	
BeSe	*B*3	5.147 7(5)	
MgSe	hex.	4.145(5); 6.723(8)	
CaSe	*B*1	5.932(2)	
SrSe	*B*1	6.243 2(12)	
BaSe	*B*1	6.593(16)	
EuSe	*B*1	6.192	
YbSe	*B*1	5.933	
$TiSe_2$	*C*6	3.540(1); 6.008(3)	
TiSe	$B8_1$	3.571; 6.301 (B31 and rhomb. phases also reported)	
$ZrSe_3$	monocl.	5.41; 3.77; 9.45 ($\beta = 97.5°$)	340
$ZrSe_2$	*C*6	3.772(2); 6.125(5)	
U_3Se_4	$D7_3$	8.820(1)	334
U_2Se_3	$D5_8$	11.34; 4.057; 10.92	334
$ThSe_2$	*C*23	7.629; 4.435; 9.085	
Th_7Se_{12}	*D*8*k*	11.570(6); 4.23(1)	
Th_2Se_3	$D5_8$	11.57(5); 4.27(1); 11.34(5)	
ThSe	*B*1	5.889 6(2)	
USe_3	monocl.	5.652; 4.056; 10.469; 115.03°	180
VSe_2	*C*6	3.356; 6.150	
VSe	$B8_1$	3.660; 5.988	
Cr_2Se_3	hex.	6.28; 11.64 (several other structures reported)	
CrSe	$B8_1$	3.699 1(4); 6.072 1(8)	
$CrNaSe_2$	$F5_1$	3.732 3(2); 20.396(3); 120°	
WSe_2	*C*7	3.282(1); 12.96(1)	
$MnSe_2$	*C*2	6.430	
MnSe (α)	*B*1	5.46(6)	
(β)	*B*3	5.82	
(γ)	*B*4	4.12; 6.72	
$FeSe_2$	*C*18	4.804(2); 5.784(3); 3.586(2) (C2 phase also reported)	
FeSe	$B8_1$	3.632; 5.910	
	*B*10	3.775; 5.527 ($L1_0$ structure also reported)	
$CoSe_2$	*C*2	5.8593(5)	
CoSe	$B8_1$	3.613; 5.300	
$NiSe_2$	*C*2	5.973	
NiSe	$B8_1$	3.652; 5.347	
	*B*13	10.007; 3.333; 120°	
$RuSe_2$	*C*2	5.930	
$RhSe_2$	*C*2	5.963 9(6)	181
$OsSe_2$	*C*2	5.946(1)	
$PtSe_2$	*C*6	3.727 8; 5.081 3	
CuSe	*B*18	3.976; 17.243 (several other hex. and ortho. phases reported)	
Cu_2Se	*C*1	5.759(1) (Berzelianite) (numerous other structures also reported)	

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Metalloids, *etc.*: B, C, P, N, Po, Te, Se, S—*continued*			
Ag_2Se	cubic	5.006(1) (ortho. phase also reported)	
ZnSe	*B*3	5.663 (B4 structure also reported)	
CdSe	*B*3	6.04(3)	
	*B*4	4.299 5; 7.012 0	
$CdCr_2Se_4$	$H1_1$	10.742 ('normal' $H1_1$-type)	182
HgSe	*B*3	5.65 (Tiemannite)	175
GaSe	hex.	3.754(1); 15.945(1)	63,
	hex.	several phases reported	184, 185
	hex.	3.755 3; 31.990(10)	63,
Ga_2Se_3	*B*3	5.446 (several other structures reported)	184
In_2Se_3	hex.	16.06; 19.28 (numerous other structures reported)	186
TlSe	*B*37	8.024; 7.18 (cubic structure also reported)	
$GeSe_2$	ortho.	6.953(1); 12.221(5); 23.04(3) (several other phases also reported)	
GeSe	*B*16	10.825; 3.833(4); 4.388(4)	336
SnSe	*B*16	11.62(1); 4.282(5); 4.334(4)	187
PbSe	*B*1	6.121 3(8)	
Sb_2Se_3	$D5_8$	11.793 8(9); 3.985 8(6); 11.647(7)	
Bi_2Se_3	*C*33	4.129; 28.60	80
S (orthorh.)	*A*16	10.464 6(1); 12.866 0(1); 24.468 0(3)	
(monocl.)	C_2^5h	11.02; 10.96; 10.96; 96.7°	
(rhomb.)	rhomb.	10.9; 4.27	
Li_2S	*C*1	5.708	
Na_2S	*C*1	6.523	
K_2S	*C*1	7.393	
Rb_2S	*C*1	7.65	
BeS	*B*3	4.863(5)	
MgS	*B*1	5.199 7(5)	
CaS	*B*1	5.700 (Oldhamite)	
SrS	*B*1	6.024(6)	
BaS	*B*1	6.387(3)	
La_2S_3	$D7_3$	8.692(3) (several other structures reported)	
Ce_2S_3	$D7_3$	8.608 4(18) (several other structures reported)	
Ce_3S_4	$D7_3$	8.625 0(5)	
CeS	*B*1	5.776	
DyS	*B*1	5.489(6)	
Ac_2S_3	$D7_3$	8.99(1)	
TiS_2	*C*6	3.402(1); 5.716(1)	
ThS_2	*C*23	7.267(3); 4.273(2); 8.615(4)	
Th_2S_3	$D5_8$	10.99(5); 3.96(3); 10.85(5)	
ThS	*B*1	5.685 1(3)	
ZrS_2	*C*6	3.663(2); 5.827(6)	
U_2S_3	$D5_8$	10.62; 3.86; 10.39	
US	*B*1	5.483	
Np_2S_3	$D5_8$	10.32(1); 3.86(5); 10.62(1)	
Pu_2S_3	$D7_3$	8.454 6(8)	
PuS	*B*1	5.53	
Am_2S_3	$D7_3$	8.434 4(3)	
VS	$B8_1$	3.360; 5.813 (several other structures reported)	
TaS_2	*C*6	3.365(2); 5.853(2) (numerous other phases reported)	
CrS	$B8_1$	3.419; 5.550	
	monocl.	3.826; 5.913; 6.089; 101.60°	
$CrNaS_2$	$F5_1$	3.554; 19.52; 120°	
$CrKS_2$	$F5_1$	3.602(6) 21.15(5); 120°	
MoS_2	*C*7	3.161; 12.295 (Molybdenite)	
WS_2	*C*7	3.154; 12.36 (Tungstenite)	
MnS_2	*C*2	6.101 6	
MnS	*B*1	5.222 6 (Alabandite)	
	*B*3	5.59	
	*B*4	3.988; 6.433	
$MnCr_2S_4$	$H1_1$	10.111	

(*continued*)

Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Metalloids, *etc.*: B, C, P, N, Po, Te, Se, S—*continued*			
FeS_2	*C*2	5.293(2) (Pyrites)	
	*C*18	4.464; 5.44; 3.39 (Marcasite)	
FeS	$B8_1$	3.445; 5.763 (numerous minor variants)	183, 189
Fe_7S_8	hex.	6.867 3(9); 17.062(2) (several other structures proposed)	
$FeKS_2$	*F*5*a*	7.079(1); 11.304(2); 5.398(1); 113.20(1)°	
$FeCr_2S_4$	$H1_1$	9.998	193
CoS_2	*C*2	5.427(2)	
Co_3S_4	$H1_1$	9.404(1) (Linnaeite)	
CoS	$B8_1$	3.44; 5.79	
Co_9S_8	$D8_9$	9.930(2)	
$CoCr_2S_4$	$H1_1$	9.923	
NiS_2	*C*2	5.676 5(1)	
Ni_3S_4	$H1_1$	9.489	
NiS	$B8_1$	3.437; 5.350	
NiS	*B*13	9.607(1); 3.143(1) (Millerite)	
Ni_6S_5	ortho.	3.254; 11.334; 16.43	
Ni_3S_2	rhomb.	5.738(2); 7.126(2); 120°	
Ni_2FeS_4	$H1_1$	9.465	
RuS_2	*C*2	5.607(3)	
RhS_2	*C*2	5.607(4)	
PdS	*B*34	6.429(2); 6.611(2)	
OsS_2	*C*2	5.620(2)	
PtS_2	*C*6	3.543 2; 5.038 8	
PtS	*B*17	3.470 1(3); 6.109 2(8) (Cooperite)	
CuS	*B*18	3.760(1); 16.20(4) (Covellite)	
Cu_2S	$C1_b$	5.564 (several other structures also reported)	
Cu_3VS_4	$H2_4$	5.391(5) (Sulvanite)	
$CuFeS_2$	$E1_1$	5.286 4(8); 10.410 2(8) (Chalcopyrite)	
$CuFe_2S_3$	ortho.	6.231(2); 11.117(6); 6.467(1) (Cubanite)	
Cu_5FeS_4	tetr.	10.94; 21.88 (several other structures also reported)	
$CuCo_2S_4$	$H1_1$	9.472(1) (Carrollite)	
Ag_2S	monocl.	4.231; 6.930; 9.526; 125.48° (several other phases also reported)	
$AgFeS_2$	$E1_1$	5.66; 10.30	
ZnS	*B*3	5.382 9 (Zincblende) (low temp. mod.)	
	*B*4	3.822 6(3); 6.260 5(15) (Würtzite) (high temp. mod.)	
$ZnAl_2S_4$	$H1_1$	10.009 3(10) (<~1 000°C)	
	*B*4	3.743; 6.089; >~1 000°C	
CdS	*B*4	4.100; 6.650 (Greenockite)	
	*B*1	5.32	
$CdCr_2S_4$	$H1_1$	10.242	
HgS	*B*3	5.851 4(3) (Metacinnabarite)	
	*B*9	4.147 9(7); 9.496 0(15) (Cinnabar)	
$HgCr_2S_4$	$H1_1$	10.239(1)	
Al_2S_3	*B*4	3.579(4); 5.829(7)	
Ga_2S_3 (α)	*B*3	5.153; < 550°C	
(β)	*B*4	3.682(1); 6.031(1)	
InS	ortho.	4.447(1); 10.648(2); 3.944(1)	198
In_2S_3	$H1_1$	10.728 (various other structures also reported)	
In_2CaS_4	$H1_1$	10.774(9)	
In_2MgS_4	$H1_1$	10.719(6)	
In_2MnS_4	$H1_1$	10.739 6(5)	
In_2FeS_4	$H1_1$	10.618(3)	
In_2CoS_4	$H1_1$	10.646(5)	
In_2NiS_4	$H1_1$	10.518(1)	
In_2HgS_4	$H1_1$	10.812(7)	
TlS	*B*37	7.77(1); 6.79(1)	
Tl_2S	rhomb.	12.20; 18.20; 120°	
SiS_2	*C*42	9.545(3); 5.564(2); 5.552(2)	
GeS_2	*C*44	11.691(8); 22.41(3); 6.68(1)	

(*continued*)

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Table 6.1 STRUCTURES OF METALS, METALLOIDS AND THEIR COMPOUNDS—*continued*

Element or compound	*Structure type*	*Lattice constants, remarks*	*Additional Refs.*
Metalloids, *etc.*: B, C, P, N, Po, Te, Se, S—*continued*			
GeS	*B*16	10.449(9); 3.653(4); 4.306(4)	
$(Fe, Ge)Cu_2S_4$	$E1_1$	5.332(1); 10.531(1)	
SnS_2	*C*6	3.66; 5.96	
SnS	*B*16	11.32(1); 4.050(4); 4.242(4) (several other structures also reported)	
$SnCu_2FeS_4$	tetr.	5.449(2); 10.757(3)	
PbS	*B*1	5.914 3(4) (Galena)	
$PbSnS_2$	*B*16	11.35; 4.05; 4.29 (Teallite)	
AsS	*B*1	7.153(4); 9.994(6); 12.986(6); 120.6° (Realgar)	
As_2S_3	*D*5*f*	4.256(2); 9.577(4); 12.191(5); 109.76(8)° (Orpiment)	
CoAsS	*C*2	5.576(2) (Cobaltite)	
NiAsS	$F0_1$	5.690 (Gersdorffite)	
CuAsS	ortho.	11.35; 5.456; 3.749	
Sb_2S_3	$D5_8$	11.380(5); 3.829(2); 11.189(5) (Stibnite)	
FeSbS	monocl.	6.02; 5.93; 6.02; 67.87° (Gudmundite)	
NiSbS	$F0_1$	5.933(3) (Ullmannite)	
$CuSbS_2$	$F5_6$	6.008(10); 3.784(10); 14.456(30) (Wolfsbergite)	
Bi_2S_3	$D5_8$	11.31; 3.98; 11.13 (Bismuthite)	
$NaBiS_2$	*B*1	5.75(2)	
$KBiS_2$	*B*1	6.00(1)	
$CuBiS_2$	$F5_6$	6.142 6(3); 3.918 9(4); 14.528 2(7) (Emplecite)	
$AgBiS_2$	*B*1	5.648	
	hex.	4.07(2); 19.06(5)	
Bi_2Te_2S	rhomb.	4.14; 29.45; 120°	

Table 6.2 STRUCTURAL DETAILS

See note at end of references for sources of these data.

*A*1 (Cu type)

Cubic: O_h^5–$Fm3m$; $a = 3.614\,91$; $A = 4$
Co-ordinates: 4Cu(O_h): 000, $\frac{1}{2}\frac{1}{2}0$ ↷

*A*2 (W type)

Cubic: O_h^9–$Im3m$; $a = 3.163\,79$; $A = 2$
Co-ordinates: 2W(O_h): 000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$

*A*3 (Mg type)

Hexagonal: D_{6h}^4–$P6_3/mmc$; $a = 3.209\,44$, $c = 5.210\,76$; $A = 2$
Co-ordinates: 2Mg(D_{3h}): $\frac{2}{3}\frac{1}{3}0$; $\frac{1}{3}\frac{2}{3}\frac{1}{2}$

*A*4 (Diamond type)

Cubic: O_h^7–$Fd3m$; $a = 3.566986(3)$; $A = 8$
Co-ordinates: 8C(T_d): 000; $\frac{1}{2}\frac{1}{2}0$; ↷; $\frac{1}{4}\frac{1}{4}\frac{1}{4}$; $\frac{3}{4}\frac{3}{4}\frac{1}{4}$; ↷

*A*5 (Tin type)

Tetragonal: D_{4h}^{19}–$I4/amd$; $a = 5.830\,8$, $c = 3.181\,0$; $A = 4$
Co-ordinates: 4Sn(D_{2d}): 000; $\frac{1}{2}\frac{1}{2}\frac{1}{2}$; $\frac{1}{2}0\frac{1}{4}$; $0\frac{1}{2}\frac{3}{4}$

*A*6 (In type)

Tetragonal: D_{4h}^{17}–$F4/mmm$; $a = 3.252\,2$, $c = 4.950\,7$; $A = 4$
Co-ordinates: 4In(D_{4h}): 000; $\frac{1}{2}\frac{1}{2}0$; ↷

A7 (As type)

Rhombohedral: D_{3d}^5–$R\bar{3}m$
Co-ordinates:
Rhombohedral (I), 2As(C_{3v}): $\pm(xxx)$
Rhombohedral (II), 8As(C_{3v}): $(000; \frac{1}{2}\frac{1}{2}0;$ ↷$) \pm (xxx)$
Hexagonal (III), 6As(C_{3v}): $(000; \frac{2}{3}\frac{1}{3}\frac{1}{3}; \frac{1}{3}\frac{2}{3}\frac{2}{3}) \pm (00x)$

(*continued*)